



Combining GNNs and Sentence Encoders for Knowledge-aware Recommendations

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Integrantes:

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Pablo Utrera

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Contexto - Problema general

Los sistemas recomendadores ayudan a filtrar la sobrecarga de información en grandes catálogos.

El objetivo es sugerir ítems relevantes para cada usuario.

Modelos recientes aprovechan:

- GNNs / GCNs sobre grafos de usuarios, ítems y Knowledge Graphs.
- Sentence encoders basados en transformers para el contenido textual.



Contexto - Limitaciones

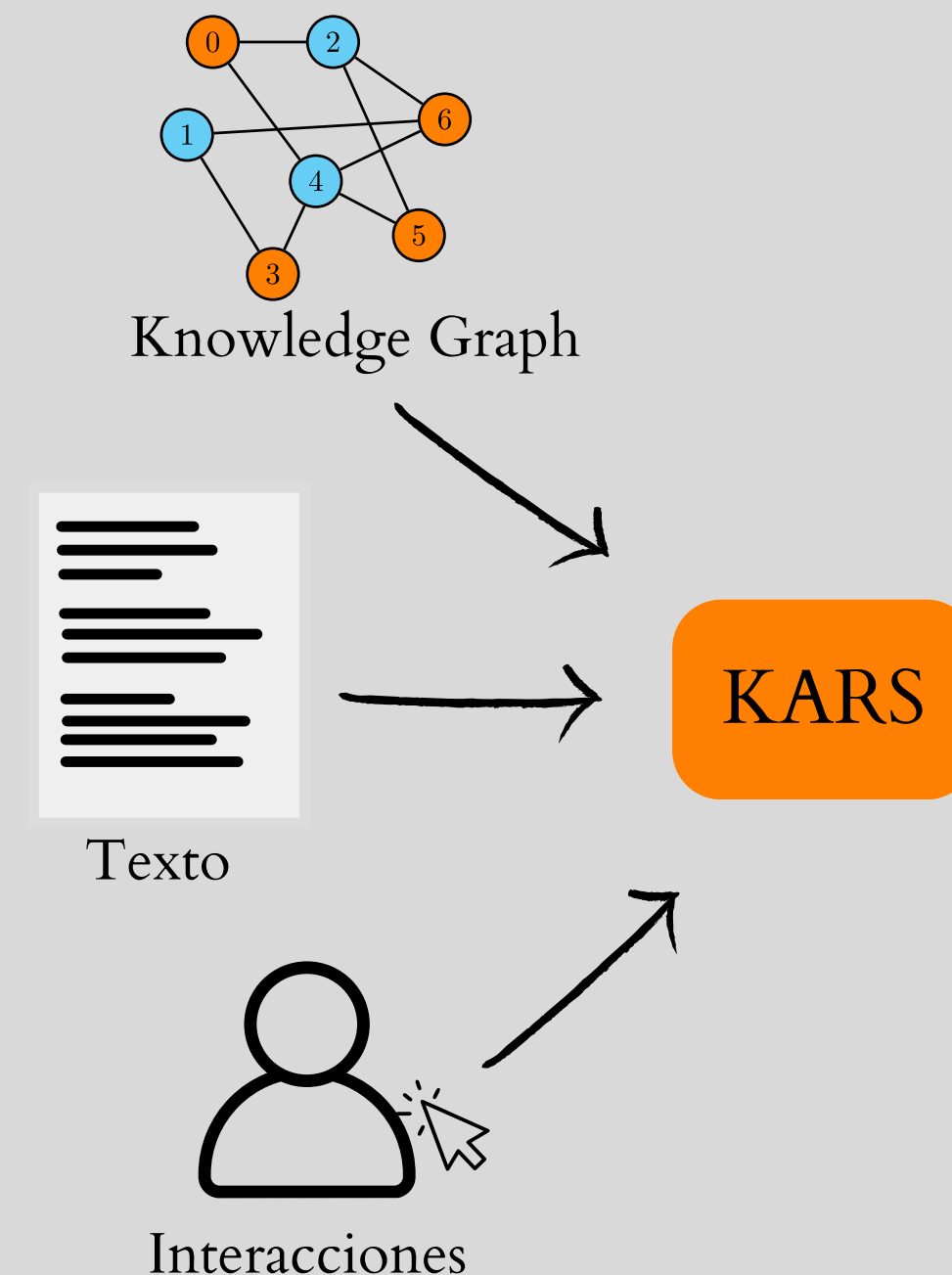
Muchos Recommender Systems usan solo una fuente de información:

- Colaborativa o texto o Knowledge Graph.

En la práctica, estas fuentes son complementarias:

- Knowledge Graph: estructura y relaciones semánticas.
- Texto: descripciones y matices.
- Interacciones: comportamiento real.

KARS busca integrar explícitamente estas fuentes en el modelo de recomendación.



Contexto - Enfoque propuesto

Estrategia para recomendaciones knowledge-aware basada en:

- Embeddings de GNNs (interacciones + Knowledge Graph).
- Embeddings de un sentence encoder (contenido textual).
- Una deep neural network con self-attention y cross-attention fusiona ambas vistas.

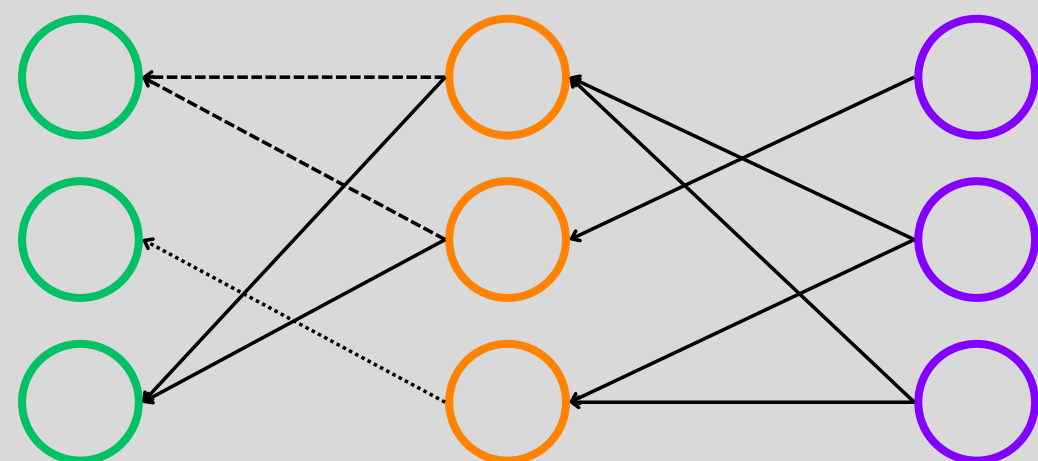
Objetivo: representaciones más ricas y mejores recomendaciones que usando una sola fuente.

Problema de Recomendación – Pregunta clave

**“Dado el
comportamiento, la
estructura y el texto,
¿qué ítems es más
probable que le interesen
ahora al usuario?”**

Problema de Recomendación

Predecir qué ítems no vistos interesan más a cada usuario utilizando:



Knowledge Graph



Historial de interacciones

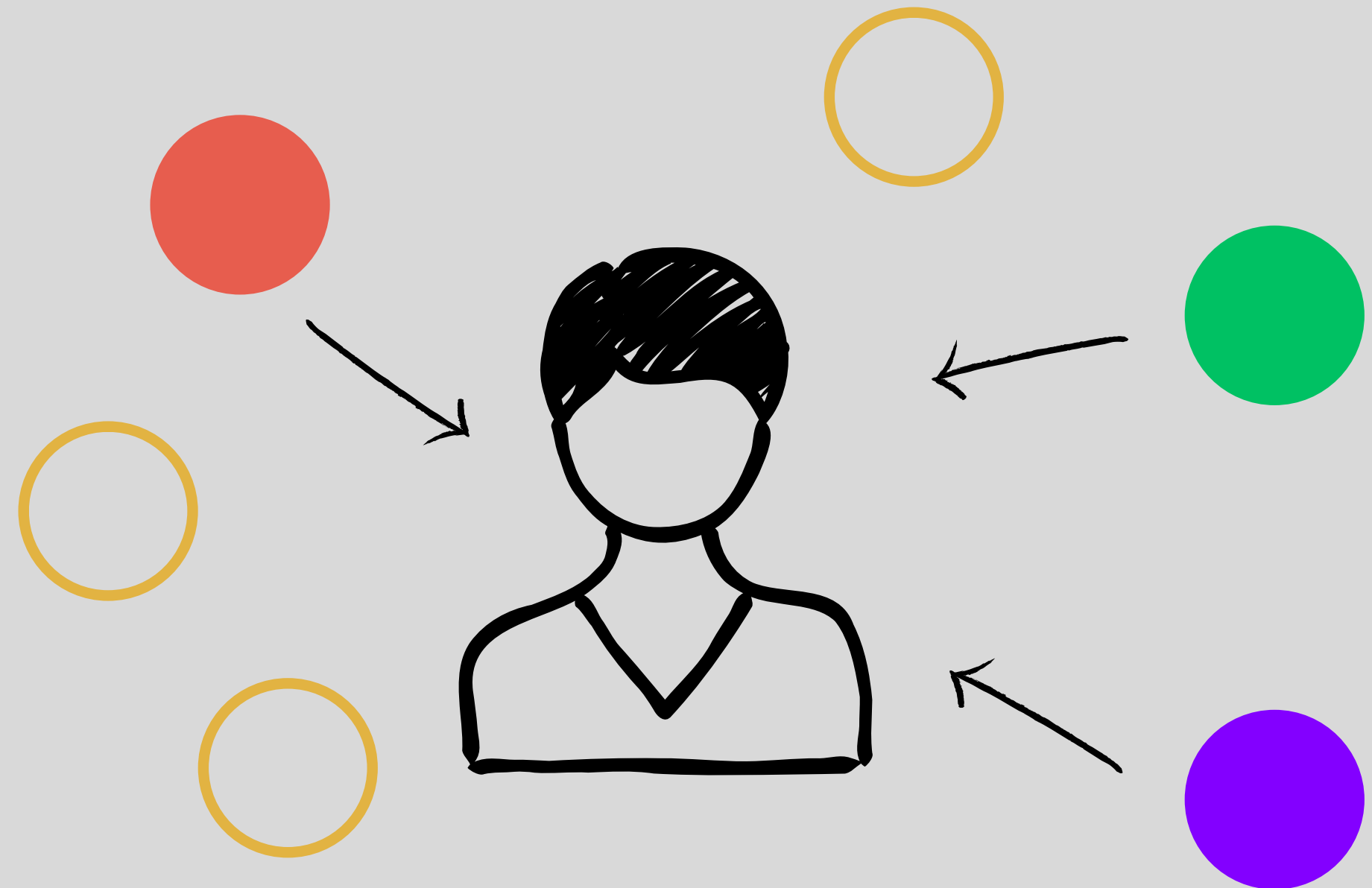


Contenido textual de los
ítems

Resultado

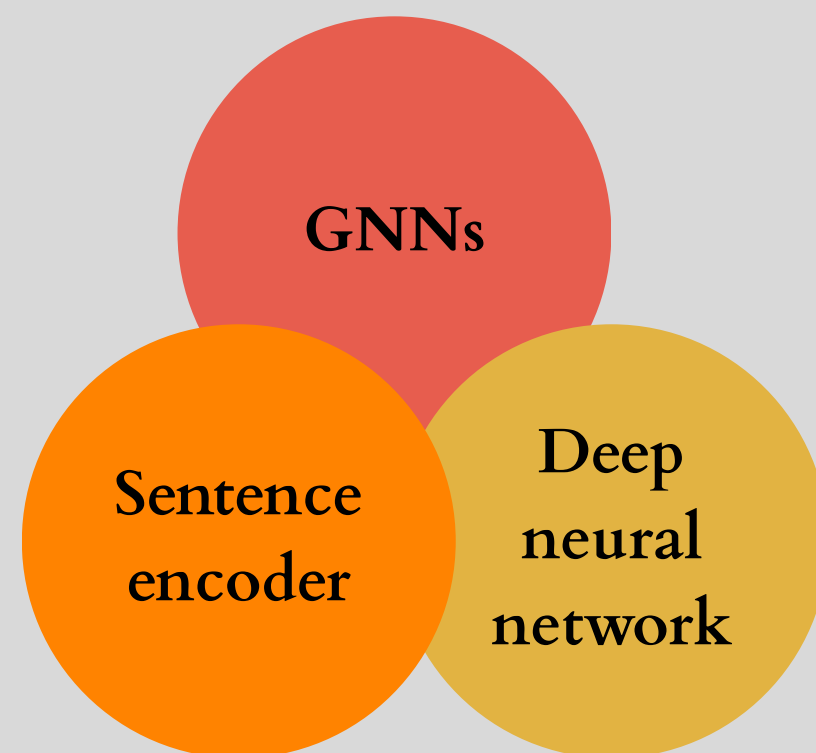
Lista de recomendación top-k personalizada.

Top 5
1. _____
2. _____
3. _____
4. _____
5. _____



¿Cómo lo enfrenta el modelo?

El enfoque combina tres ideas



El modelo

Predice el interés del usuario por
cada ítem no consumido

Genera una lista top-k ordenada
por relevancia estimada



Contribuciones

Contribuciones

1

Nueva estrategia de representación de usuarios e ítems que combina embeddings de grafo (CompGCN) y de texto (SBERT/MiniLM) usando capas de **self-attention** y **cross-attention**.

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Análisis de sensibilidad variando tipo de **GNN** y **sentence encoder**, así como **hiperparámetros** clave (dimensión, dropout, número de épocas) para estudiar la robustez del modelo.

Contribuciones

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3

Evaluación exhaustiva frente a **múltiples baselines** (MF, deep learning y KARS con GNN/GCN), donde el modelo propuesto logra **mejoras significativas** en **precisión** y **nDCG** en **recomendación top-k**.



Contribuciones

- Aportes trabajos previos



Contribuciones

- Aportes trabajos previos



KARS
Complejo

Contribuciones

- Aportes trabajos previos



KARS
Complejo



Fusión
innovadora
self-attention y
cross-attention

Contribuciones

- Aportes trabajos previos

KARS
Complejo

Fusión
innovadora
self-attention y
cross-attention

Supera
consistentemente
a modelos basados
solo en GNN o
sentence encoders

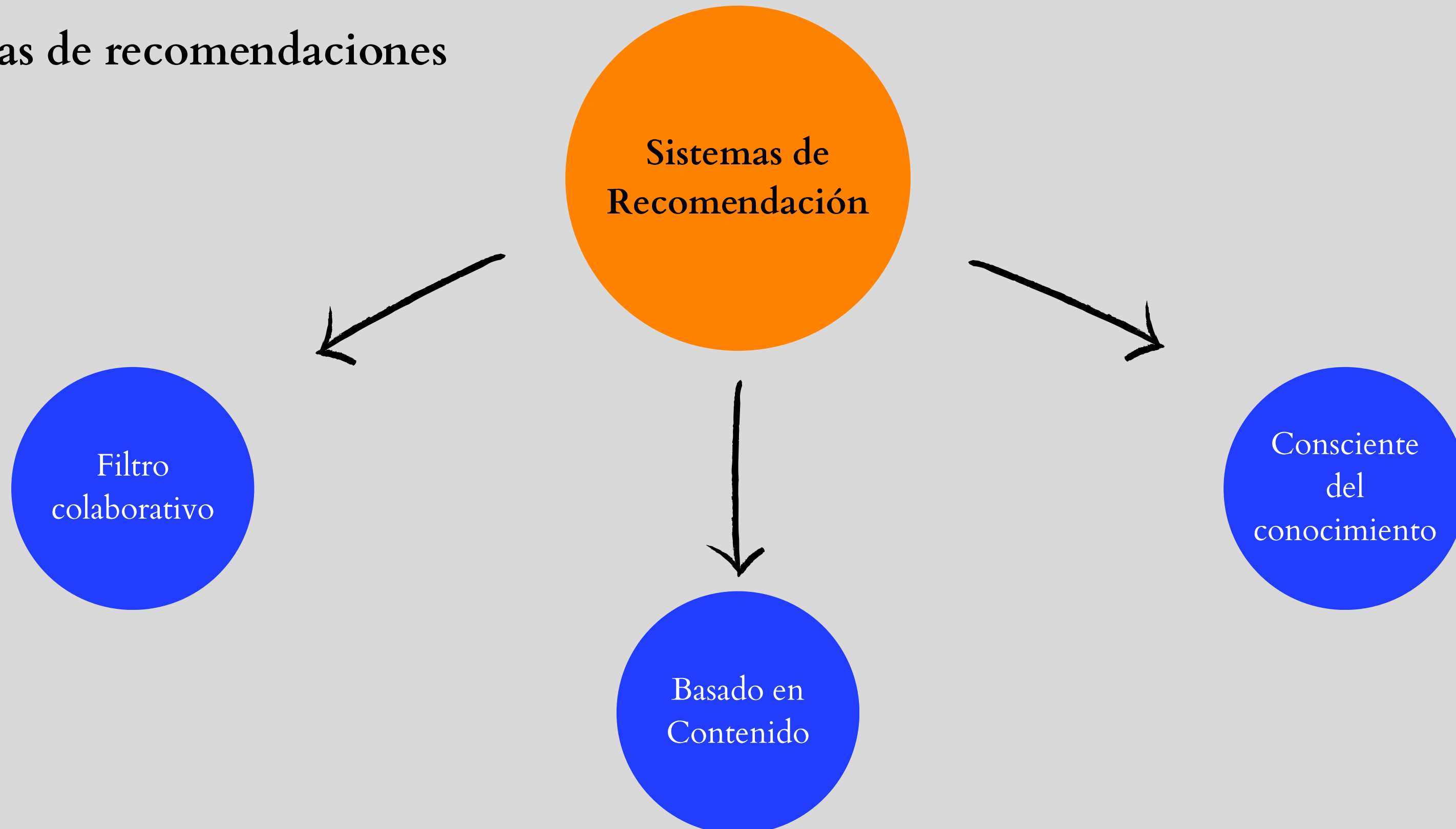


Marco Teórico

- Sistemas de recomendaciones

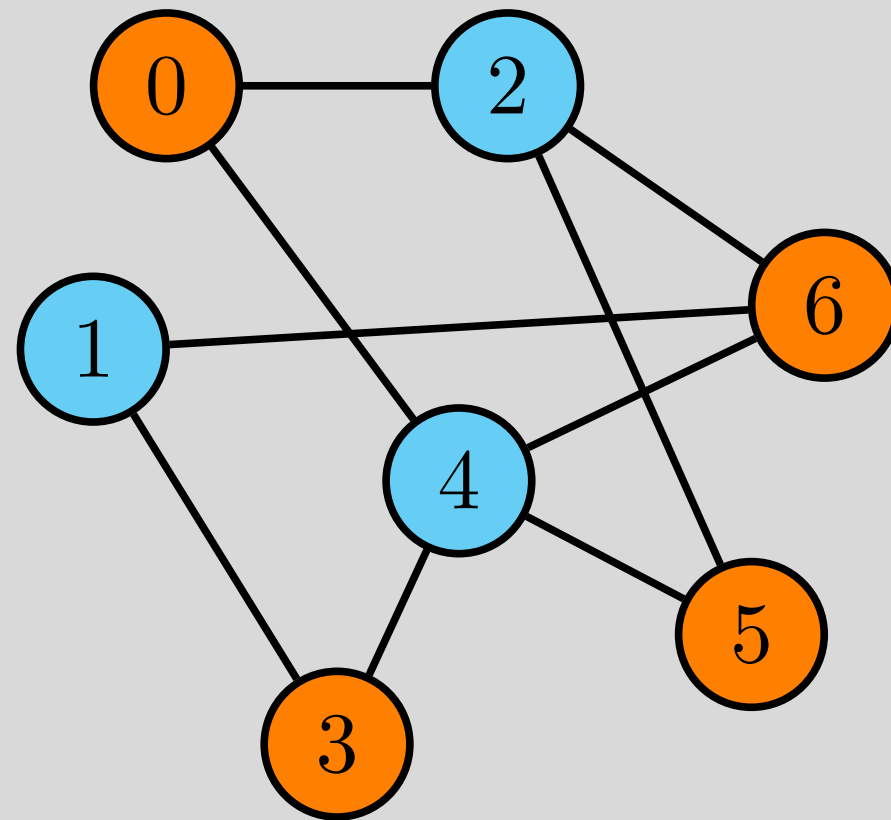
Marco Teórico

- Sistemas de recomendaciones

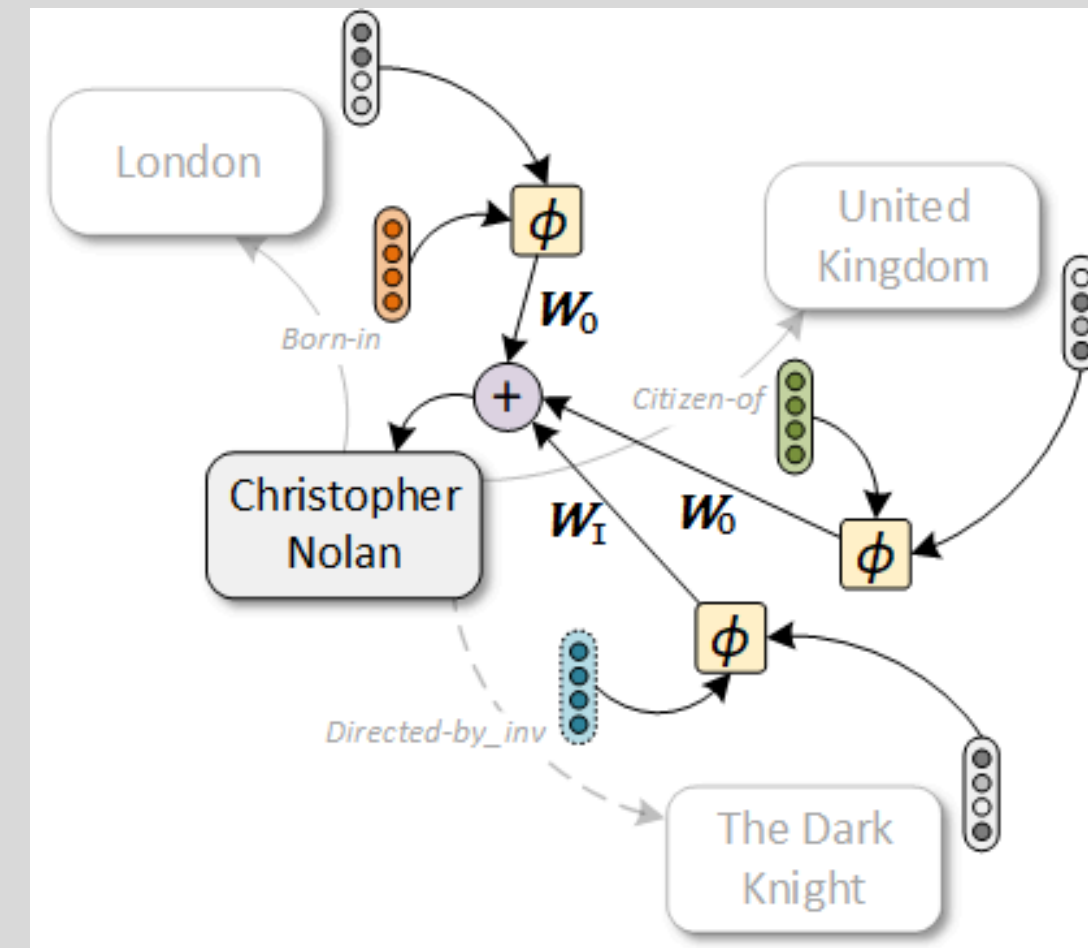


Marco Teórico

- Grafos y GNN



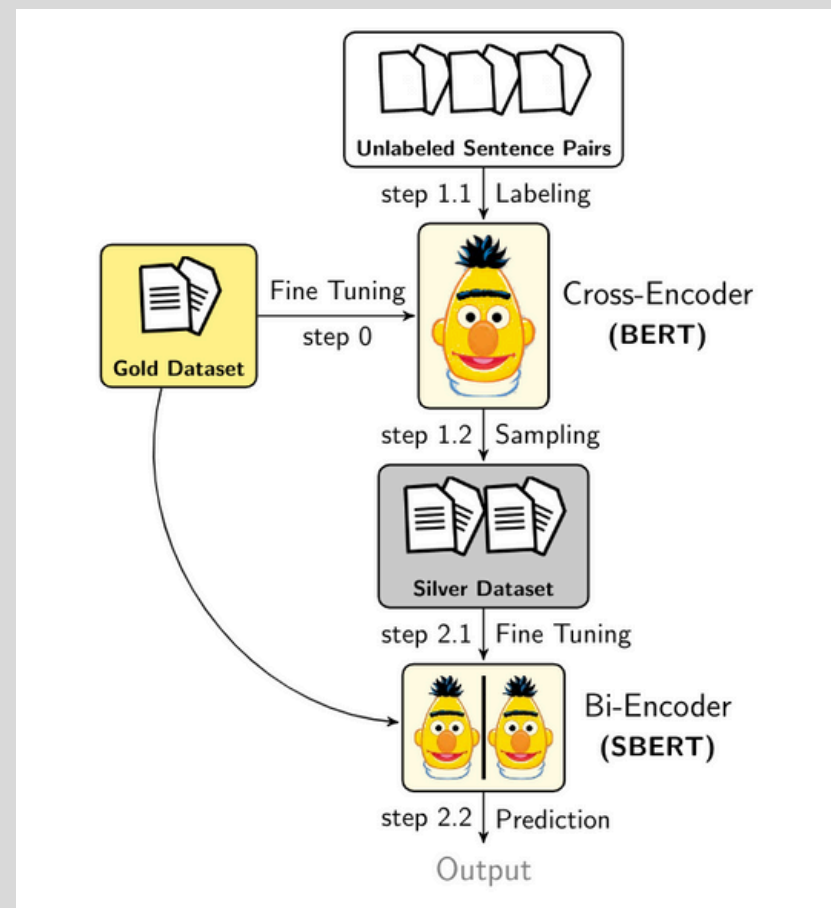
Grafo Usuario - Item



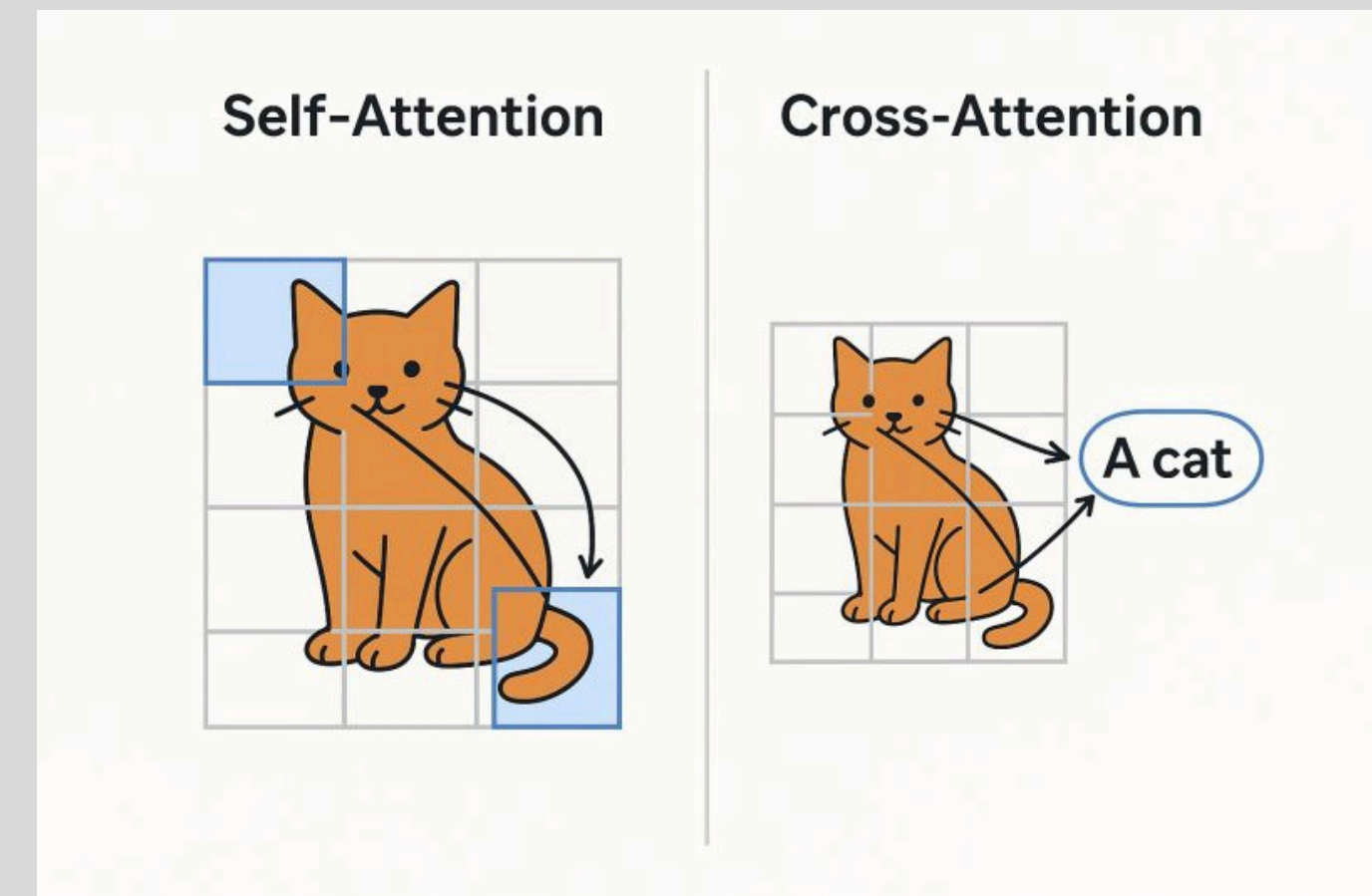
Red Neuronal
Convolutional
(ej: CompGCN)

Marco Teórico

- Sentence Encoders y Attention



Sentence Encoders
(ej: SBERT)



Self y Cross Attention

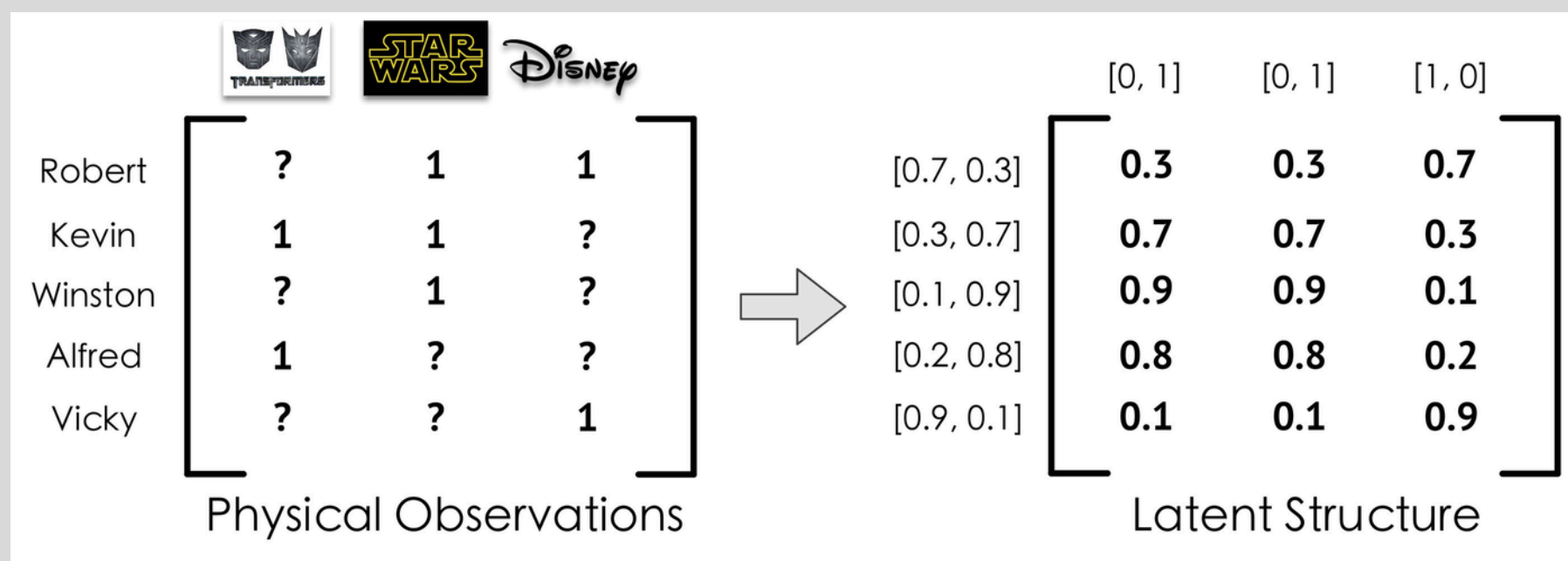
Estado del Arte



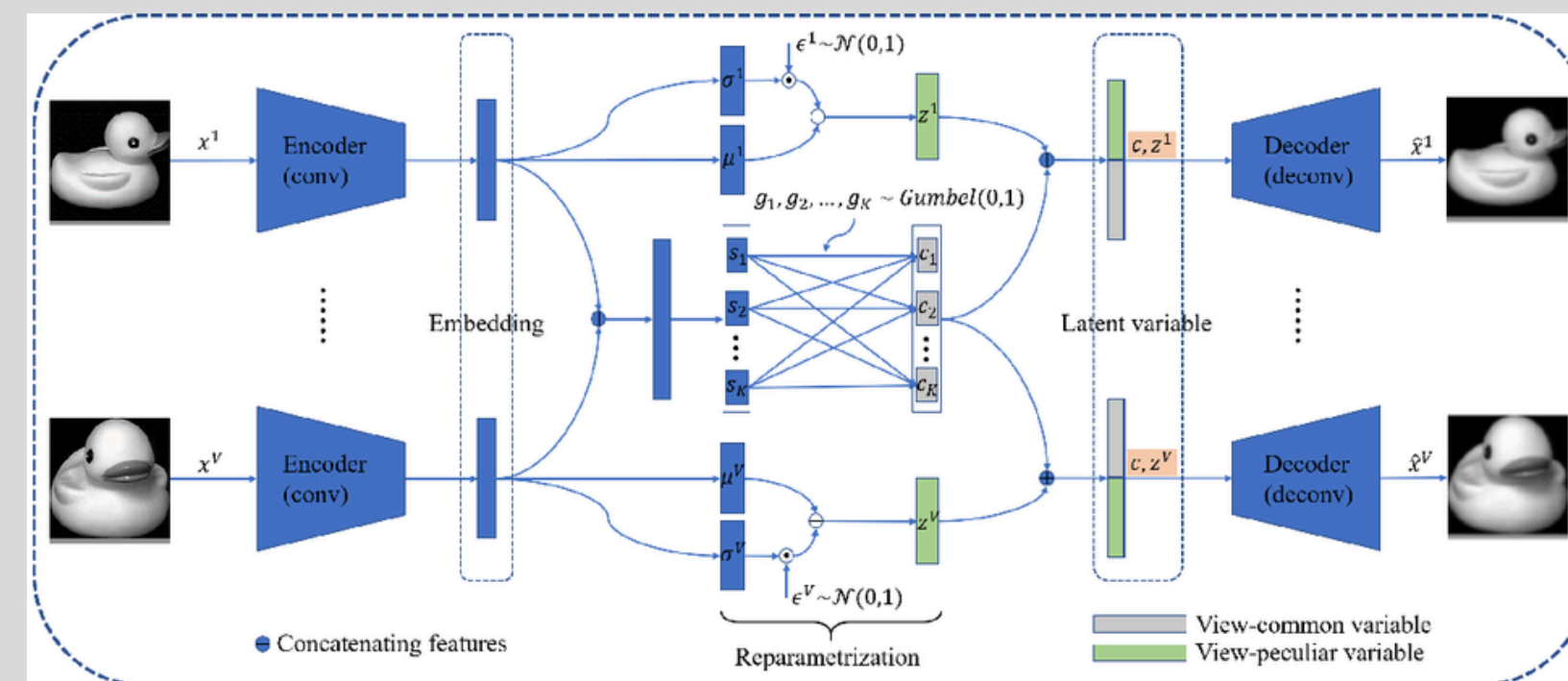
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Estado del Arte

- Modelos Sin Conocimiento



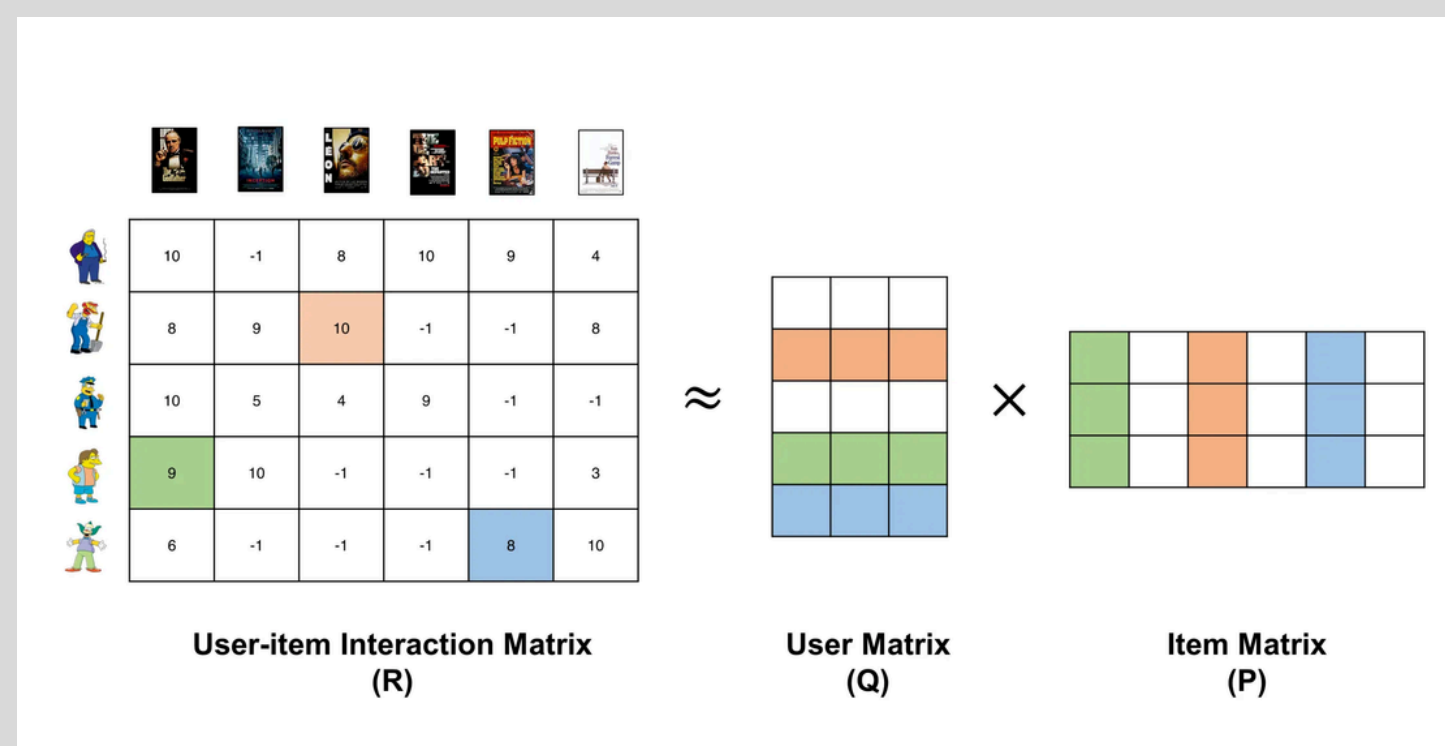
Factorization para feedback
implícito



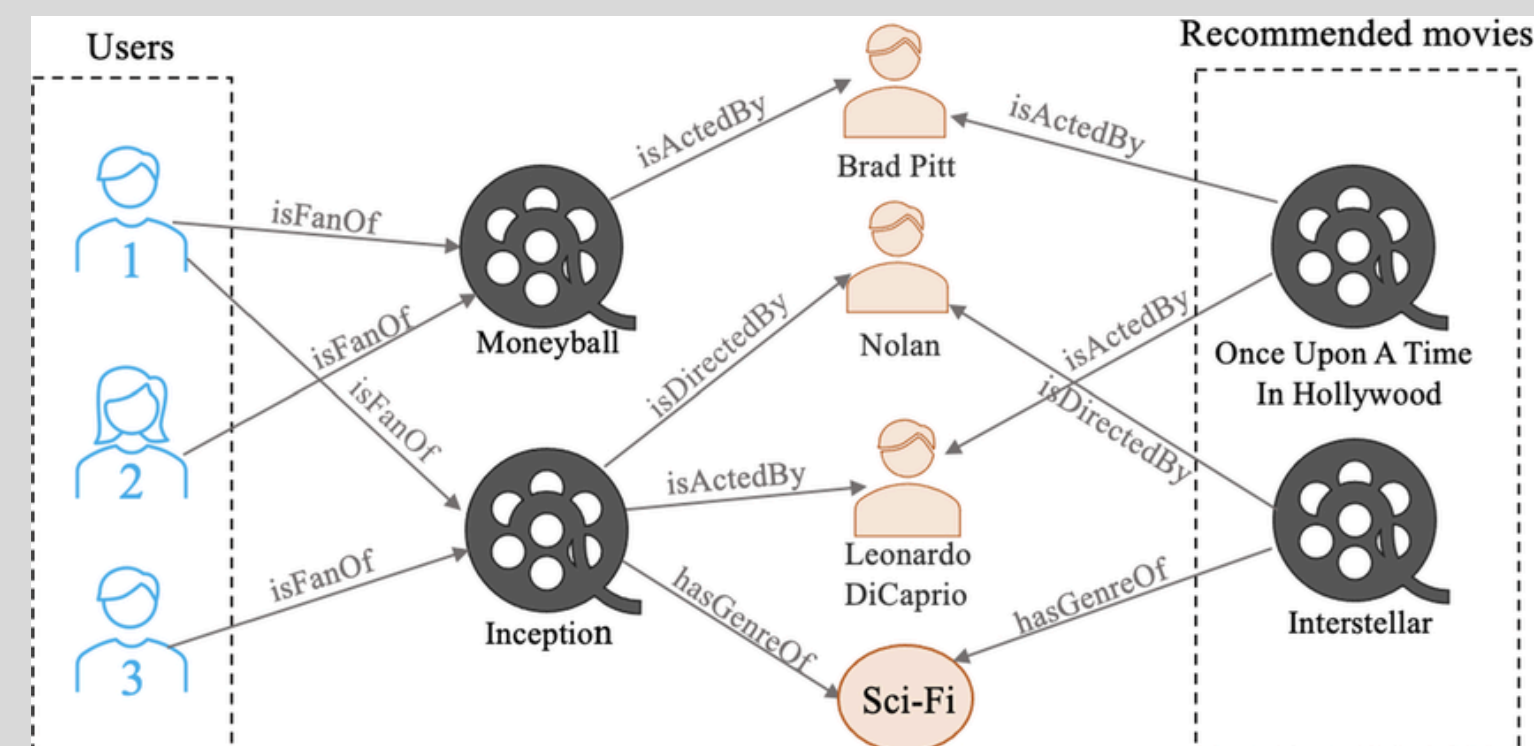
Deep sin KG
(MultiVAE)

Estado del Arte

- Modelos KARS con Grafos de Conocimiento



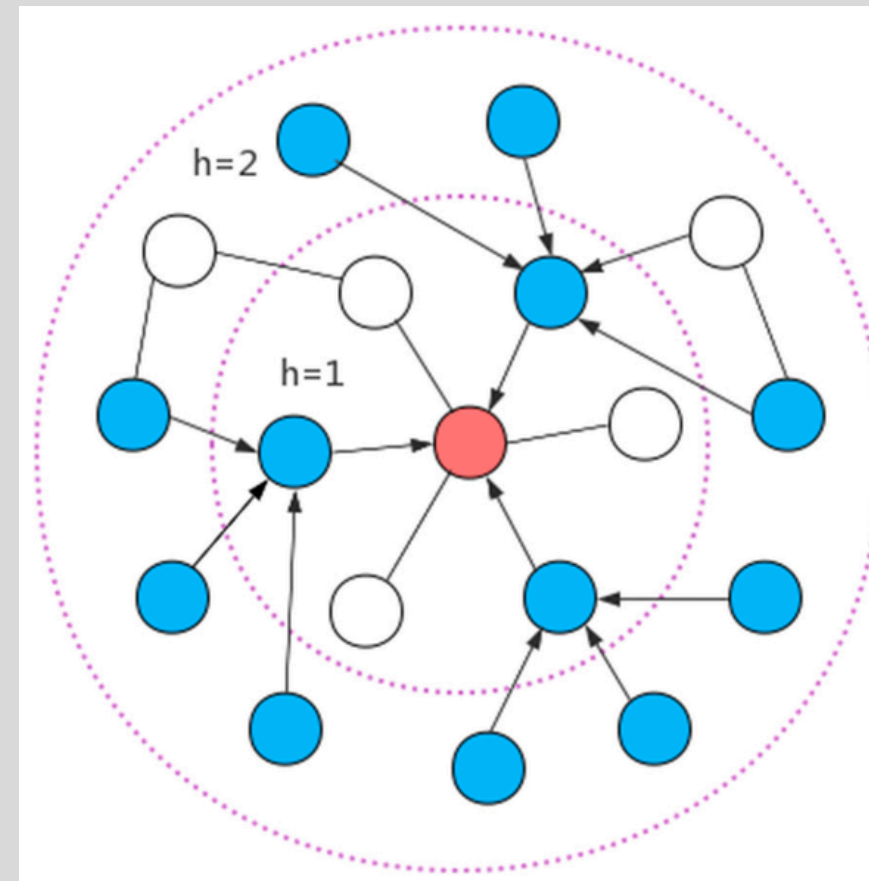
MF



KG

Estado del Arte

- Modelos KARS con Grafos de Conocimiento



GNN/GCN
(ej: KGCN)



Estado del Arte

- Limitaciones de la fusión

Concatenación Simple

$$z_i = [g_i || t_i]$$

- g_i = embedding del grafo de conocimiento (KG)
- t_i = embedding del texto (descripción, plot, etc.)
- z_i = Es el vector combinado que usará el recomendador.



Detalles de la solución

Detalles de la solución

{formulación}

Usuarios

$$\mathcal{U} = \{u_1, u_2, \dots, u_m\}$$

Ítems

$$\mathcal{I} = \{i_1, i_2, \dots, i_n\}$$

Interacciones

$$\mathcal{Y} = \mathbb{R}^{m \times n} \quad \rightarrow \quad y_{j,k} = 1$$



Detalles de la solución

{formulación}

Grafo

$$\text{IG} = \langle V, C \rangle$$

Nodos

$$|V| = |\mathcal{U}| + |\mathcal{I}|$$

Conexiones

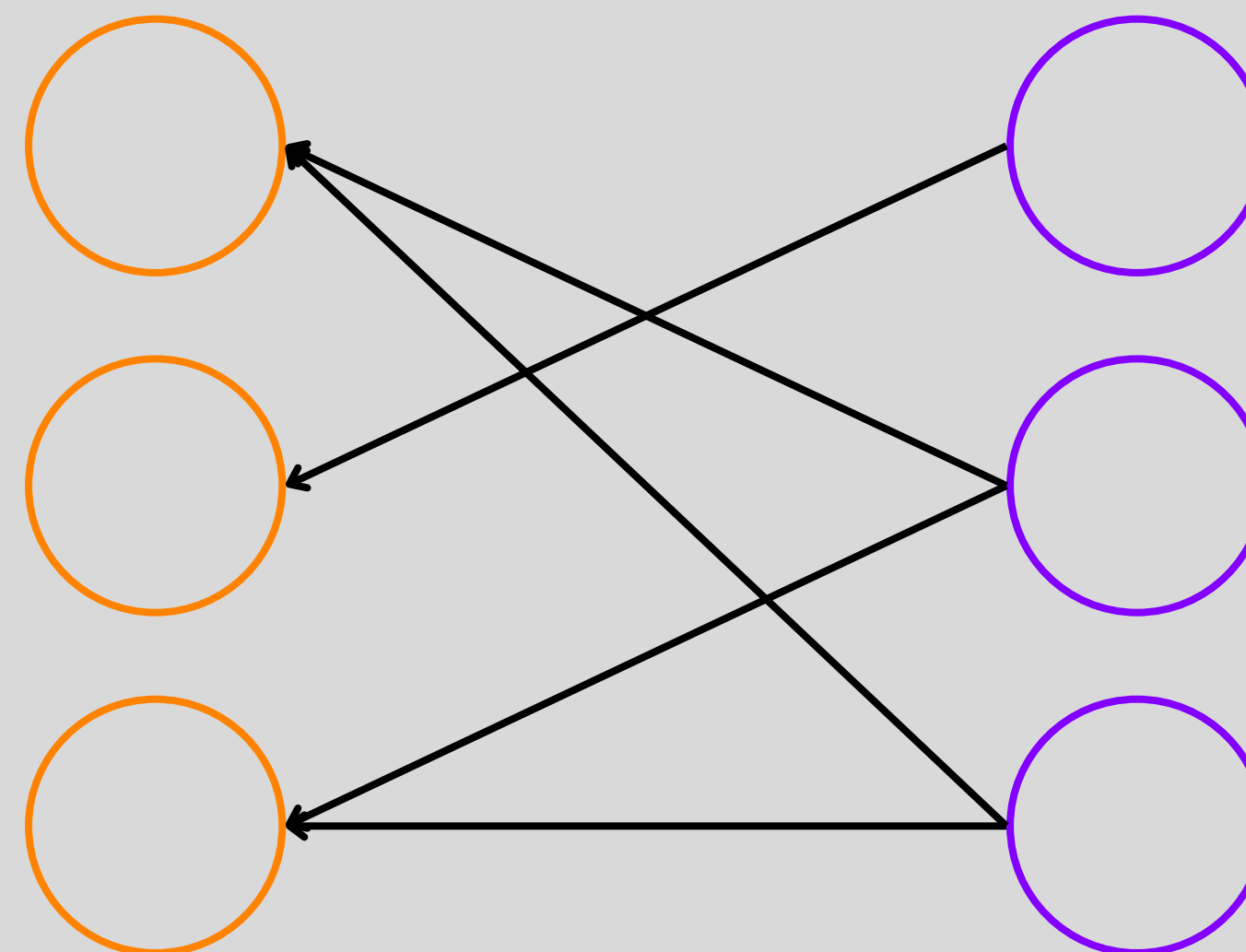
$$C$$

Detalles de la solución

{formulación}

Ítems

Usuarios



$$IG = \langle V, C \rangle$$

$$|V| = |\mathcal{U}| + |\mathcal{I}|$$

C

Detalles de la solución

{formulación}

$$\mathcal{T} = \{t_1, t_2, \dots, t_n\}$$

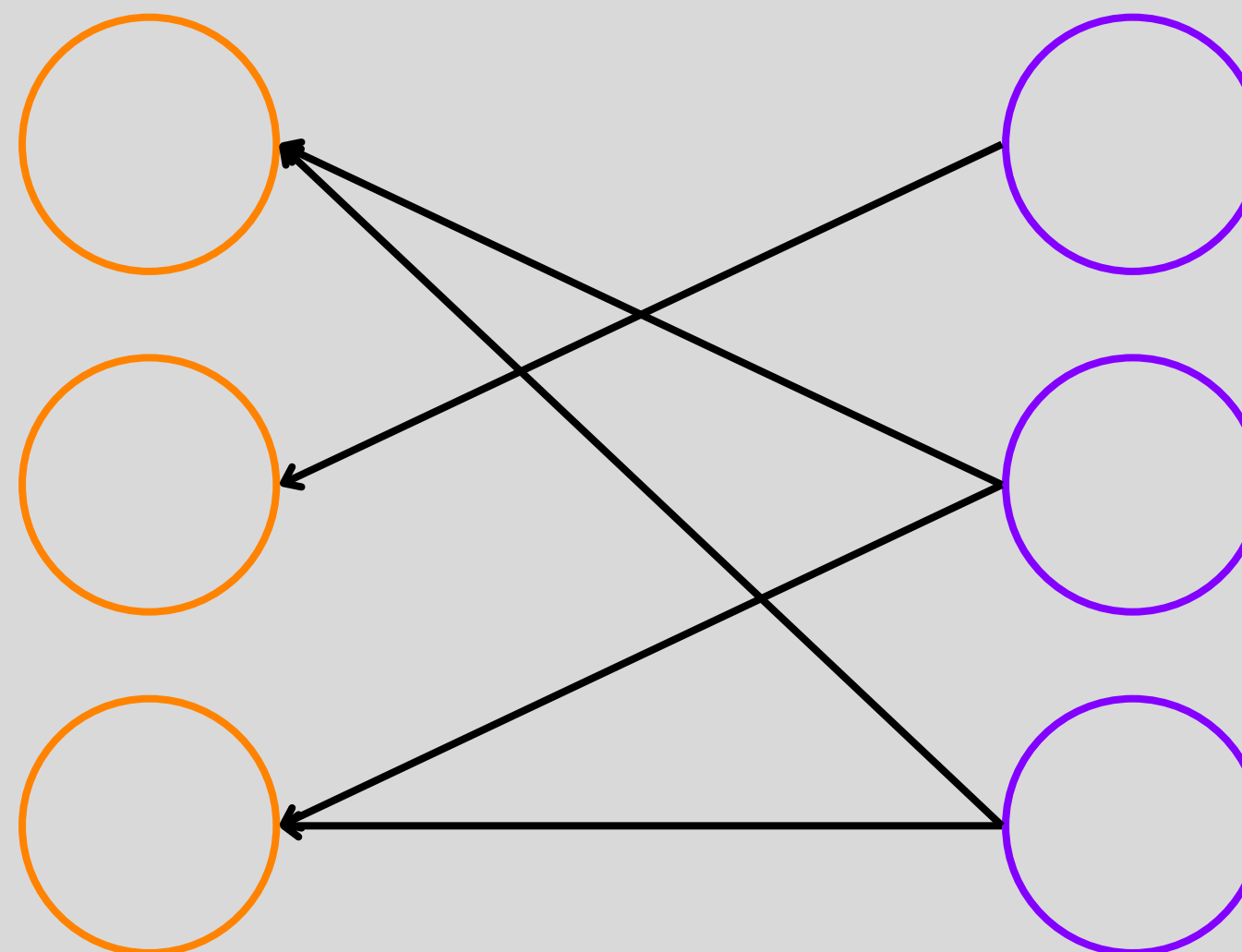
Ítems

Usuarios

Seventeen-year-old Bella Swan leaves Phoenix, Arizona and moves to Forks, a small town located on Washington state's Olympic Peninsula, to live with her father ...

Bilbo Baggins celebrates his birthday and leaves the Ring to Frodo, his heir. Gandalf (a wizard) suspects it is a Ring of Power; seventeen years...

Harry lives in a cupboard under the stairs in the house of the Dursleys, his aunt, uncle and cousin, who all treat him poorly...



Detalles de la solución

{formulación}

$$\mathcal{T} = \{t_1, t_2, \dots, t_n\}$$



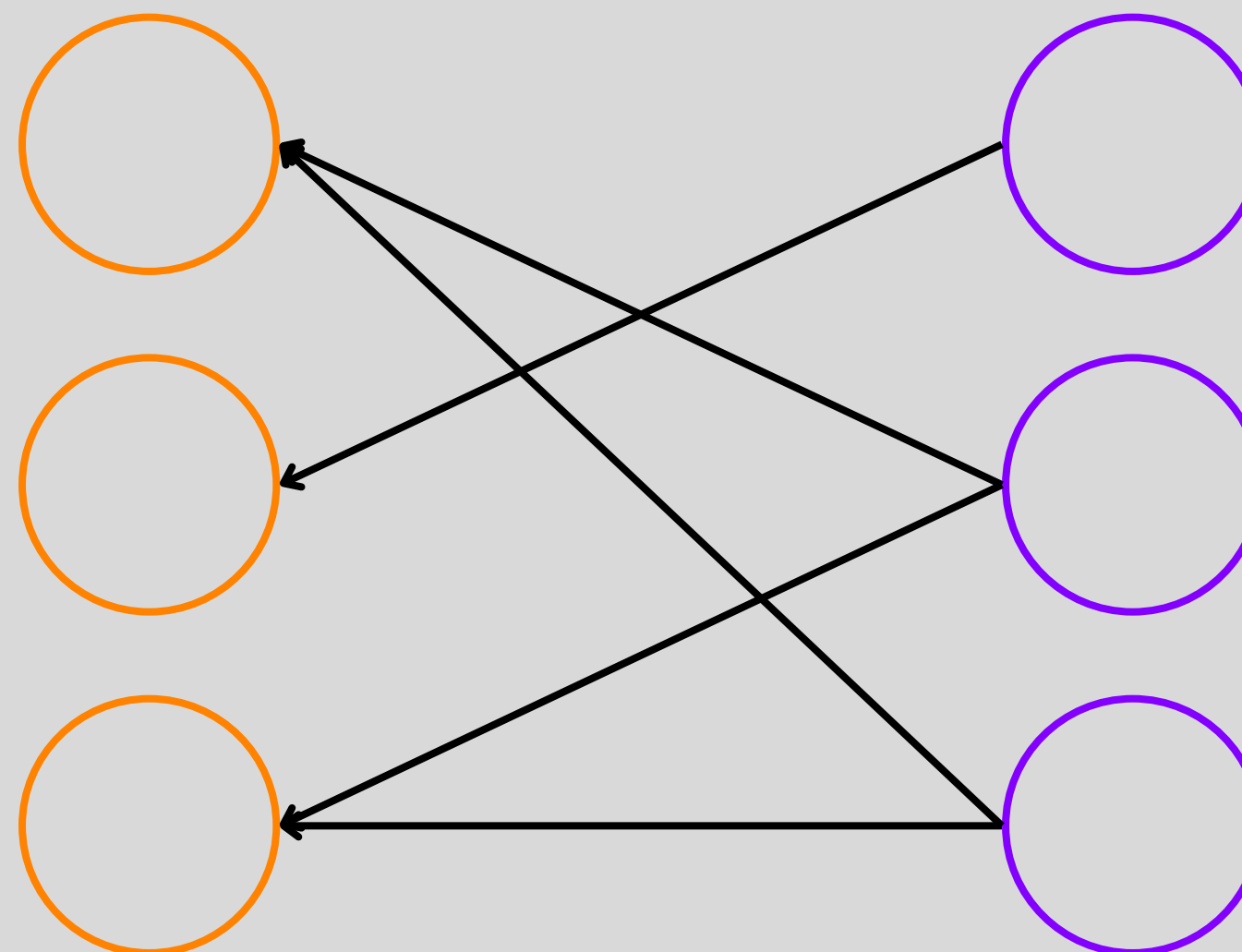
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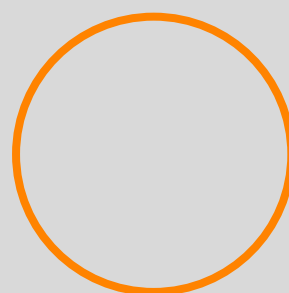
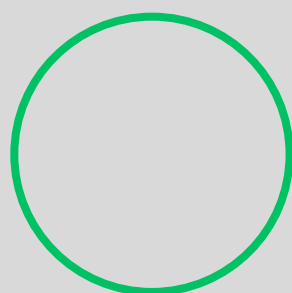
Detalles de la solución

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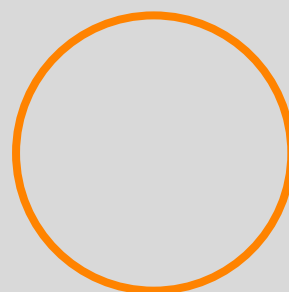
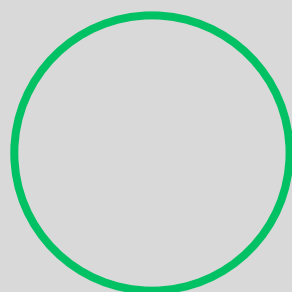
Propiedades

Ítems

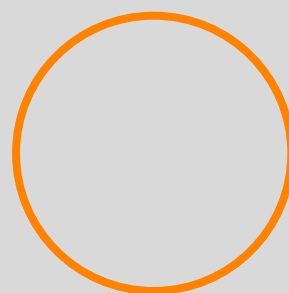
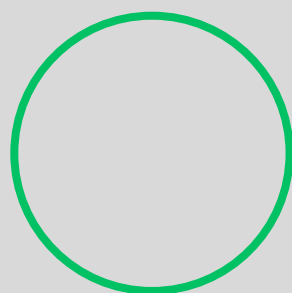
Quentin
Tarantino



James
Cameron



Sci-Fi



$$KG = \langle V', C' \rangle$$

$$|V'| = |\mathcal{P}| + |\mathcal{I}|$$

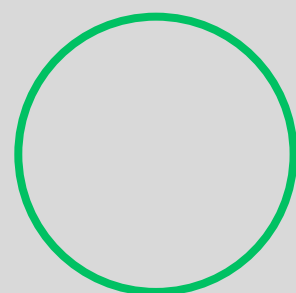
Detalles de la solución

{formulación}

Propiedades Ítems

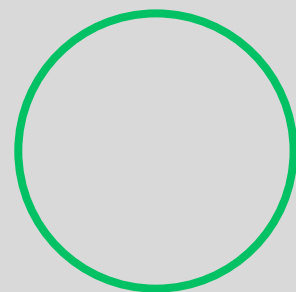
Directed by:

Quentin
Tarantino



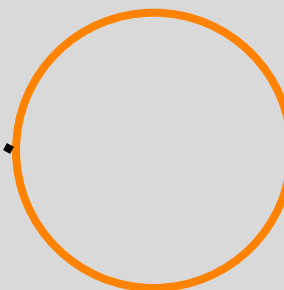
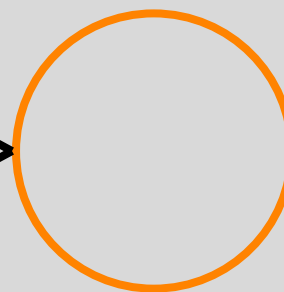
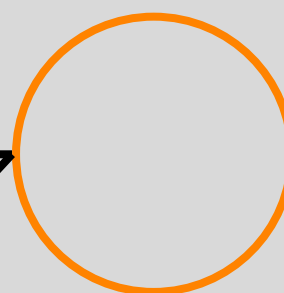
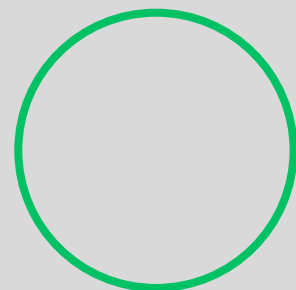
Written by:

James
Cameron



Genre:

Sci-Fi



$$KG = \langle V', C' \rangle$$

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$\mathcal{R}$

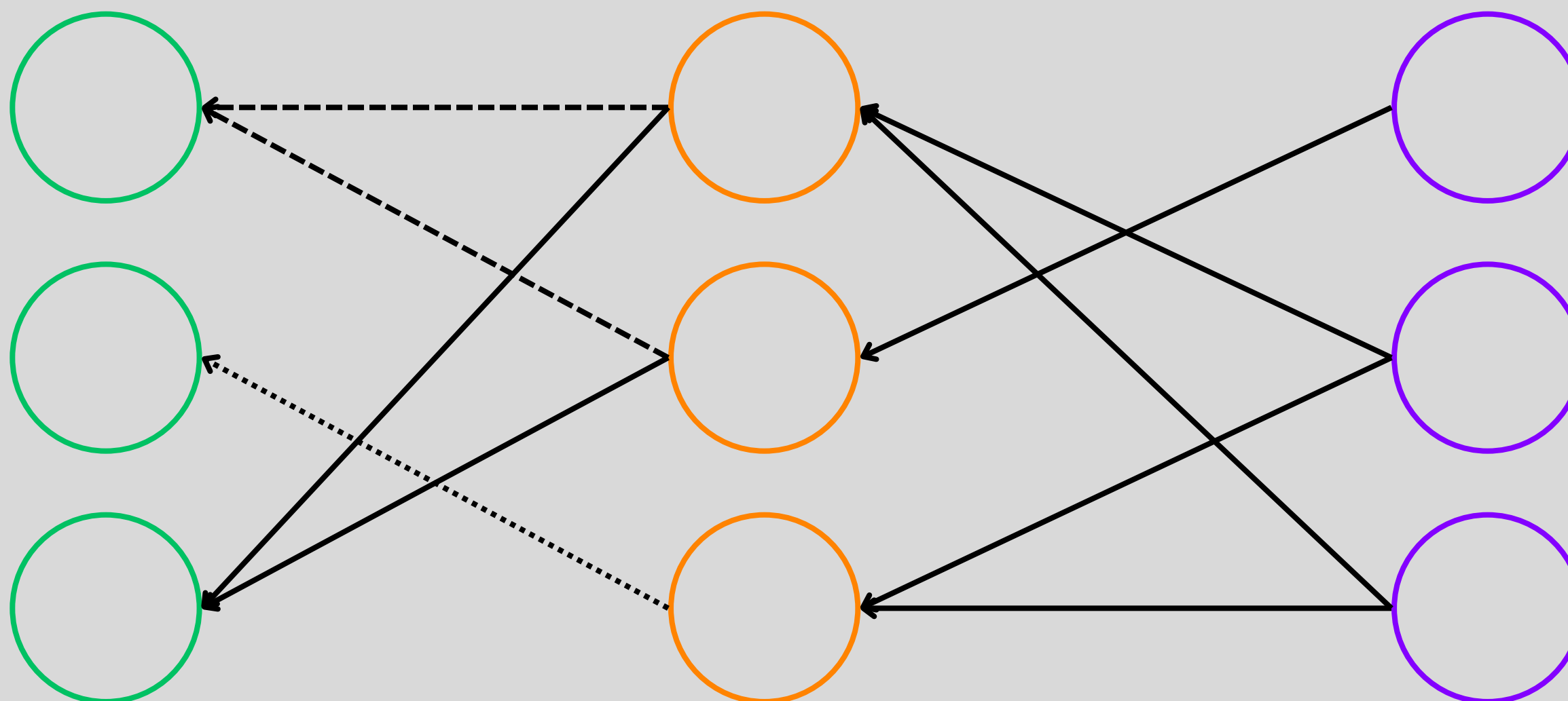
Detalles de la solución

{formulación}

Propiedades

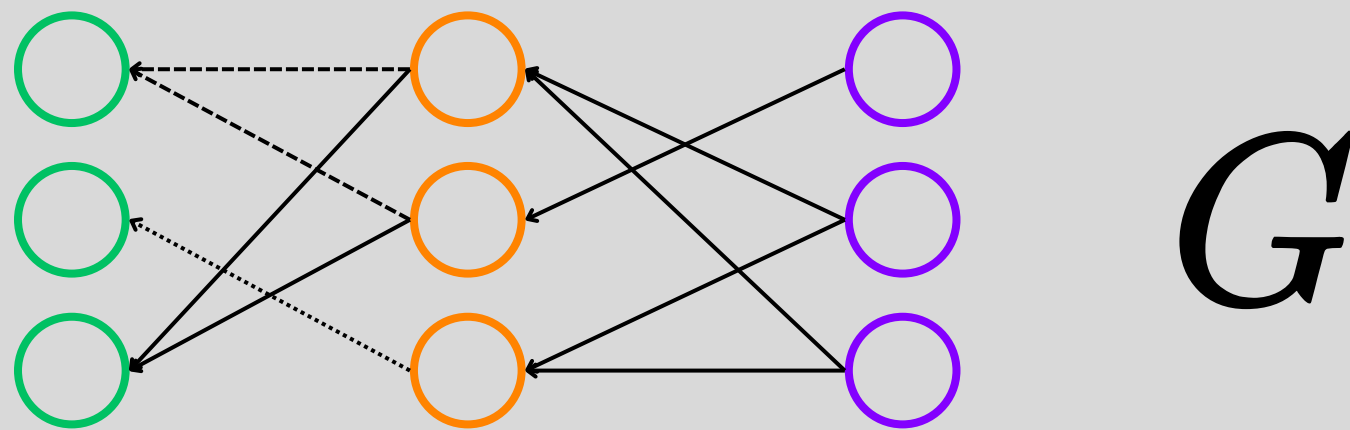
Ítems

Usuarios



Detalles de la solución

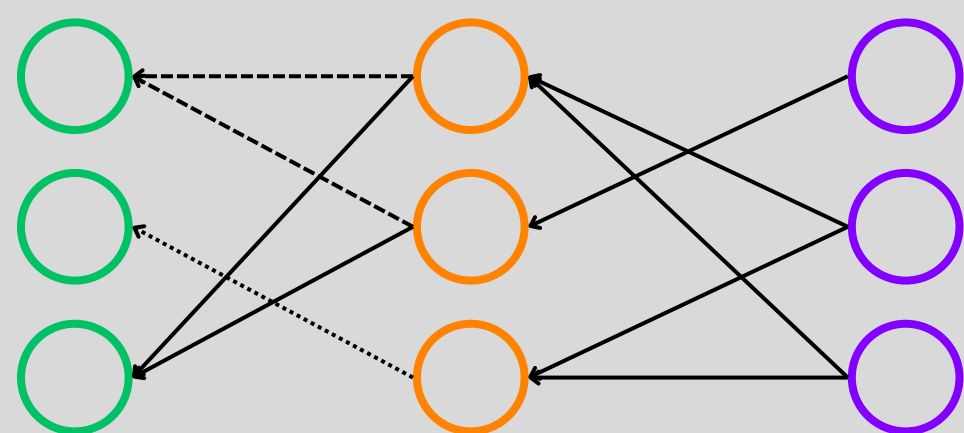
{formulación}



$$\mathcal{T} = \{t_1, t_2, \dots, t_n\}$$

Detalles de la solución

{formulación}



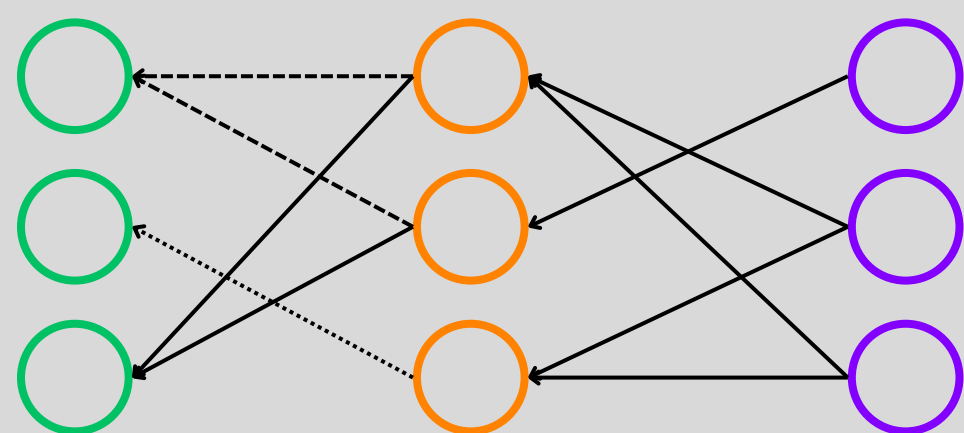
G

$\tilde{Y}_{\theta}$

$$\mathcal{T} = \{t_1, t_2, \dots, t_n\}$$

Detalles de la solución

{formulación}



G

$\tilde{Y}_{\theta}$

$y_{u,i}$

$\mathcal{T} = \{t_1, t_2, \dots, t_n\}$



Detalles de la solución

{formulación}

$$\tilde{Y}(u, i | G, \mathcal{T}, \theta) = y_{u,i}$$



Detalles de la solución

{metodología}

1. Graph Encoding
2. Sentence Encoding



Detalles de la solución

{metodología}

1. Graph Encoding
2. Sentence Encoding

Graph Encoding



{metodología}

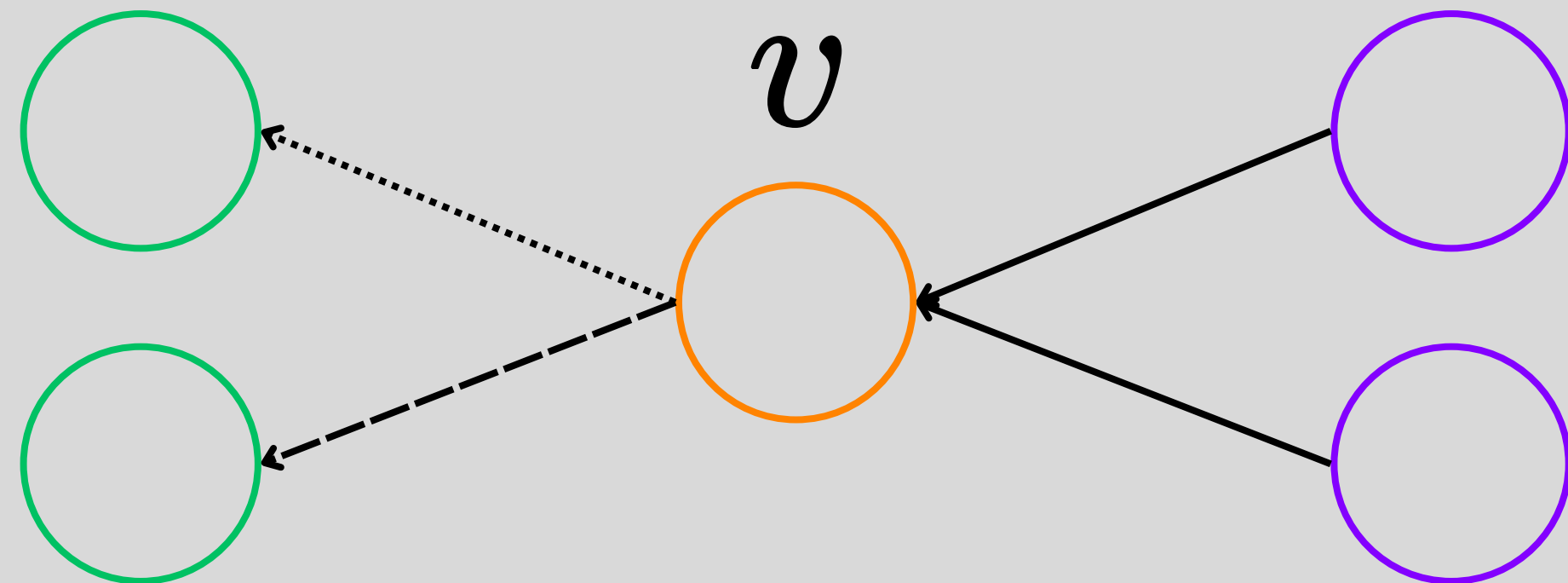
$$\vec{v} = \sum_{(u,i) \in N(v)} W_{\lambda(r)} \Phi(\vec{u}, \vec{r})$$

Graph Encoding



{metodología}

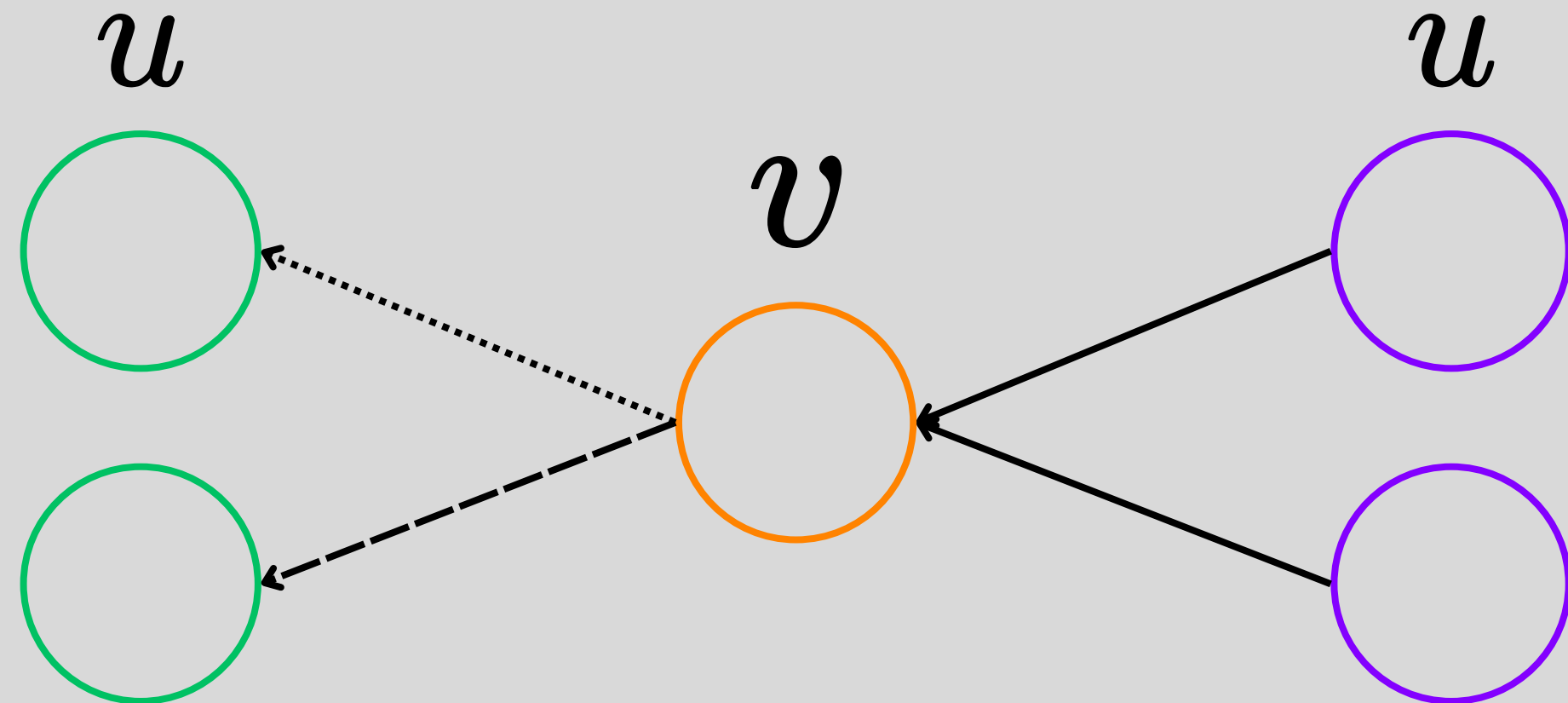
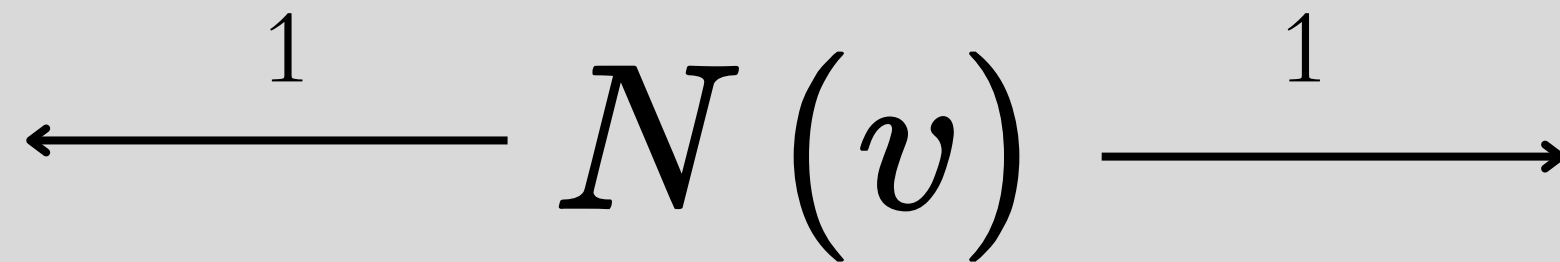
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Graph Encoding

{metodología}

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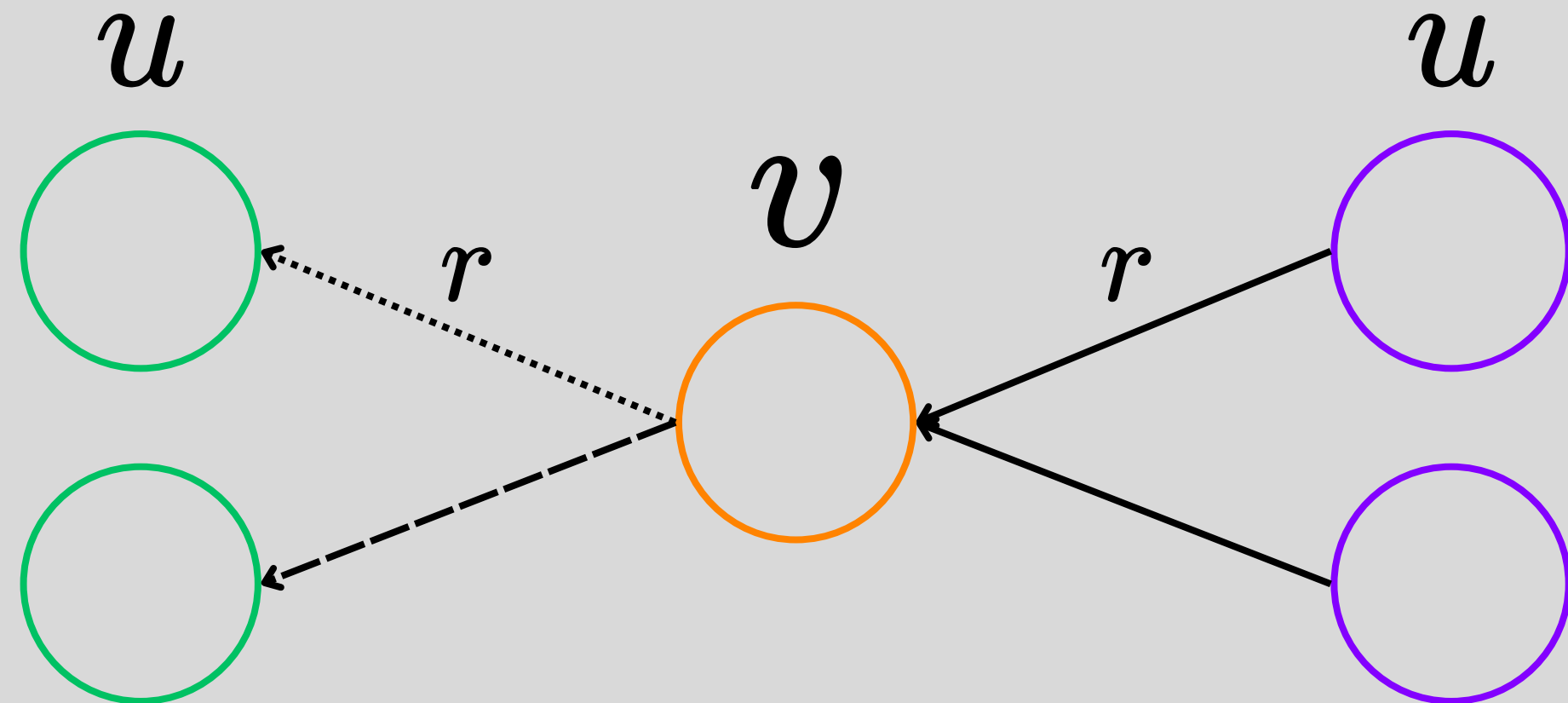


Graph Encoding

{metodología}

$$\vec{v} = \sum_{(u,i) \in N(v)} W_{\lambda(r)} \Phi(\vec{u}, \vec{r})$$

$$\xleftarrow{1} N(v) \xrightarrow{1}$$



Graph Encoding



{metodología}

$$\vec{v} = \sum_{(u,i) \in N(v)} W_{\lambda(r)} \Phi(\vec{u}, \vec{r})$$

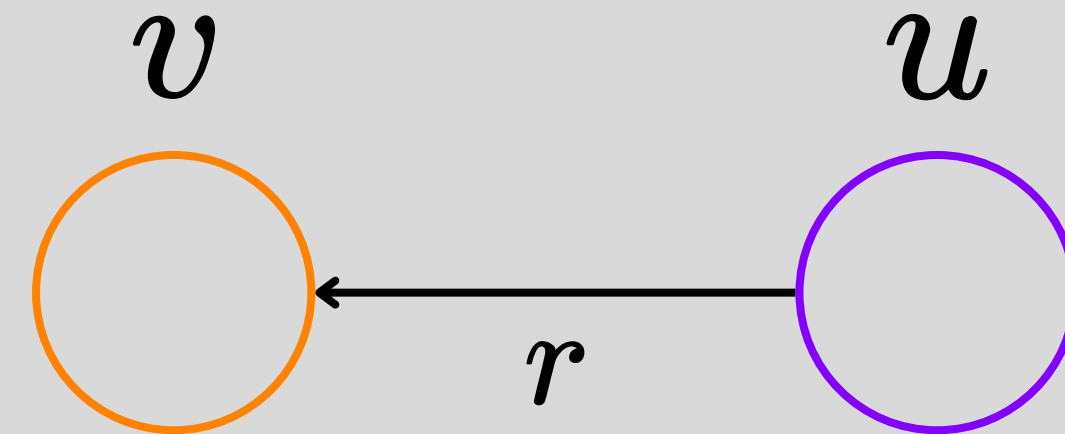
$\lambda(r)$

Graph Encoding

{metodología}

$$\vec{v} = \sum_{(u,i) \in N(v)} W_{\lambda(r)} \Phi(\vec{u}, \vec{r})$$

$\lambda(r)$

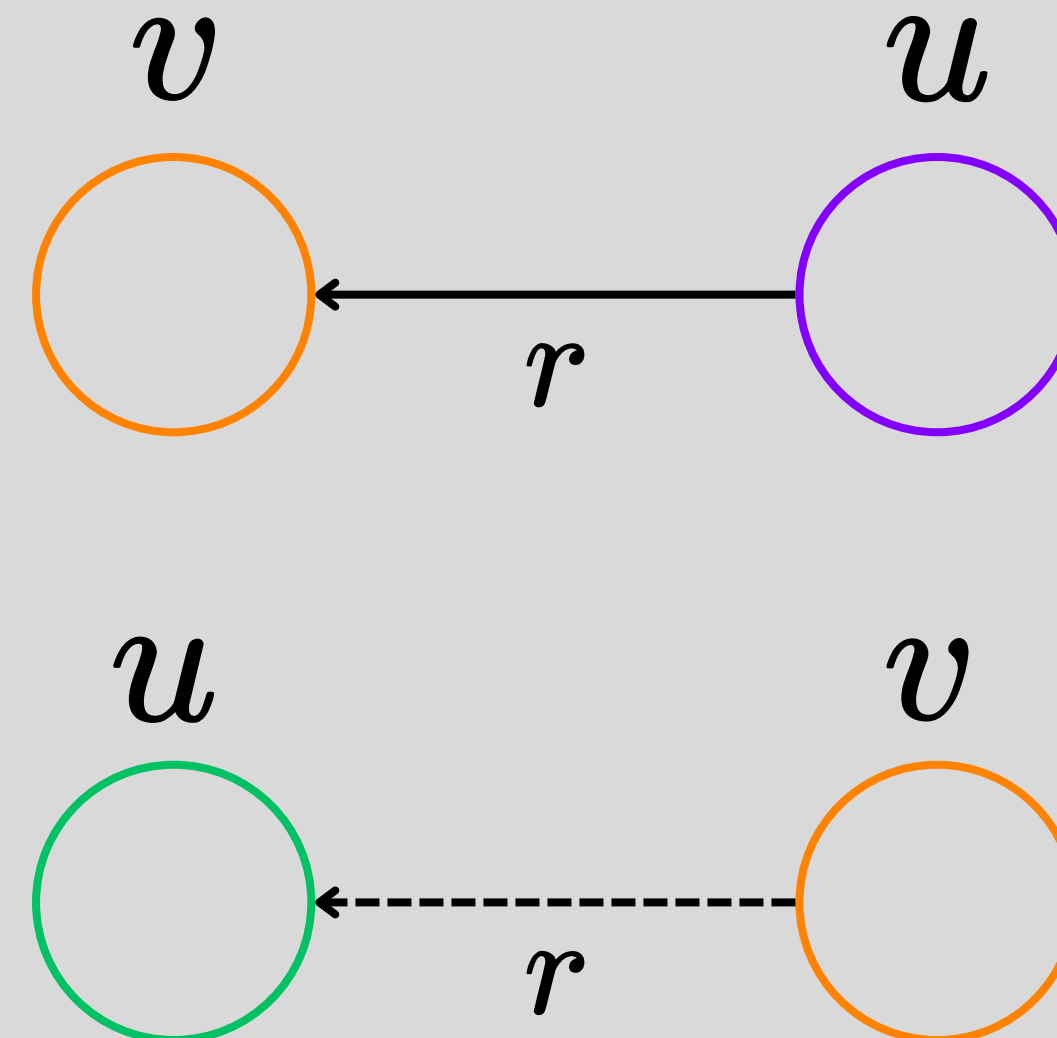


Graph Encoding

{metodología}

$$\vec{v} = \sum_{(u,i) \in N(v)} W_{\lambda(r)} \Phi(\vec{u}, \vec{r})$$

$\lambda(r)$



Graph Encoding



{metodología}

$$\vec{v} = \sum_{(u,i) \in N(v)} W_{\lambda(r)} \Phi(\vec{u}, \vec{r})$$

$\vec{r}$



Graph Encoding

{metodología}

$$\vec{v} = \sum_{(u,i) \in N(v)} W_{\lambda(r)} \Phi(\vec{u}, \vec{r})$$

$$\Phi(\vec{u}, \vec{r})$$

Graph Encoding



{metodología}

$$\vec{v} = \sum_{(u,i) \in N(v)} W_{\lambda(r)} \Phi(\vec{u}, \vec{r})$$



Detalles de la solución

{metodología}

1. Graph Encoding
2. **Sentence Encoding**

Sentence Encoding

{metodología}

MINILM: Deep Self-Attention Distillation for Task-Agnostic Compression of Pre-Trained Transformers

Wenhui Wang Furu Wei Li Dong Hangbo Bao Nan Yang Ming Zhou

Microsoft Research

{wenwan, fuwei, lidong1, t-habao, nanya, mingzhou}@microsoft.com

Abstract

Pre-trained language models (e.g., BERT (Devlin et al., 2018) and its variants) have achieved remarkable success in varieties of NLP tasks. However, these models usually consist of hundreds of millions of parameters which brings challenges for fine-tuning and online serving in real-life applications due to latency and capacity constraints. In this work, we present a simple and effective ap-

1. Introduction

Language model (LM) pre-training has achieved remarkable success for various natural language processing tasks (Peters et al., 2018; Howard & Ruder, 2018; Radford et al., 2018; Devlin et al., 2018; Dong et al., 2019; Yang et al., 2019; Joshi et al., 2019; Liu et al., 2019). The pre-trained language models, such as BERT (Devlin et al., 2018) and its variants, learn contextualized text representations by predicting words given their context using large scale text corpora, and can be fine-tuned with additional task-specific layers to



Sentence Encoding

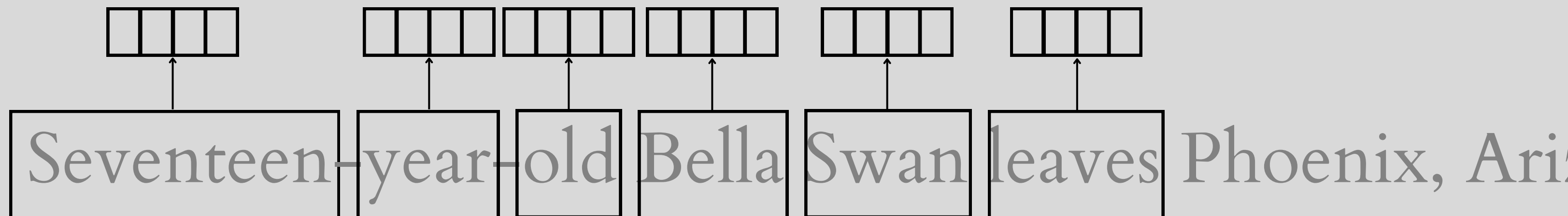
{metodología}

i_w

Sentence Encoding

{metodología}

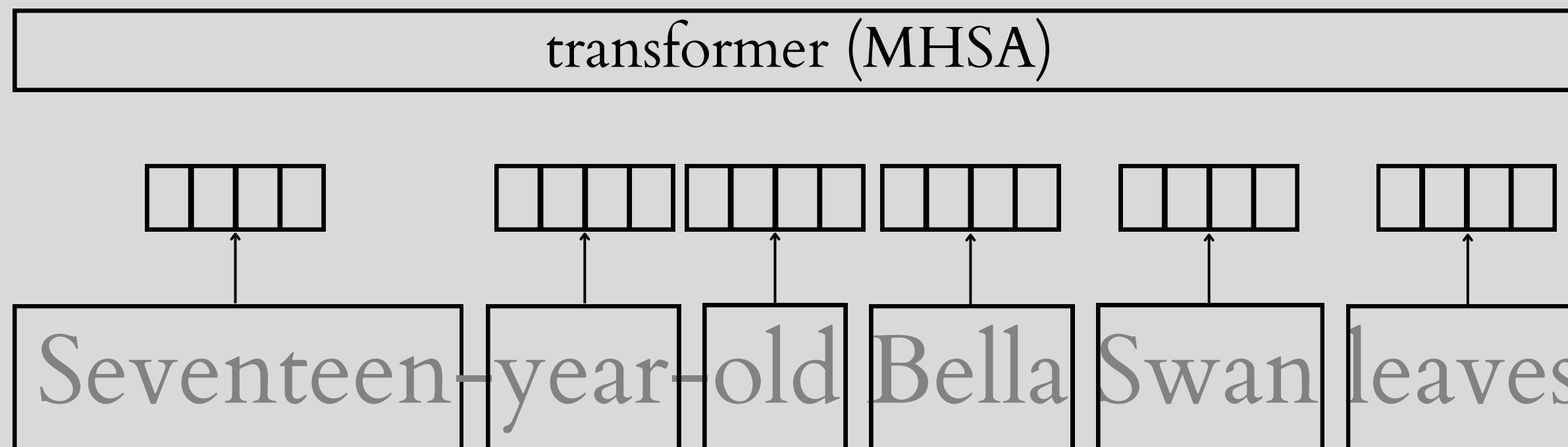
i_w



Sentence Encoding

{metodología}

i_w



Sentence Encoding

{metodología}

i_w

N {

transformer (MHSA)

...

transformer (MHSA)



Seventeen

year

old

Bella

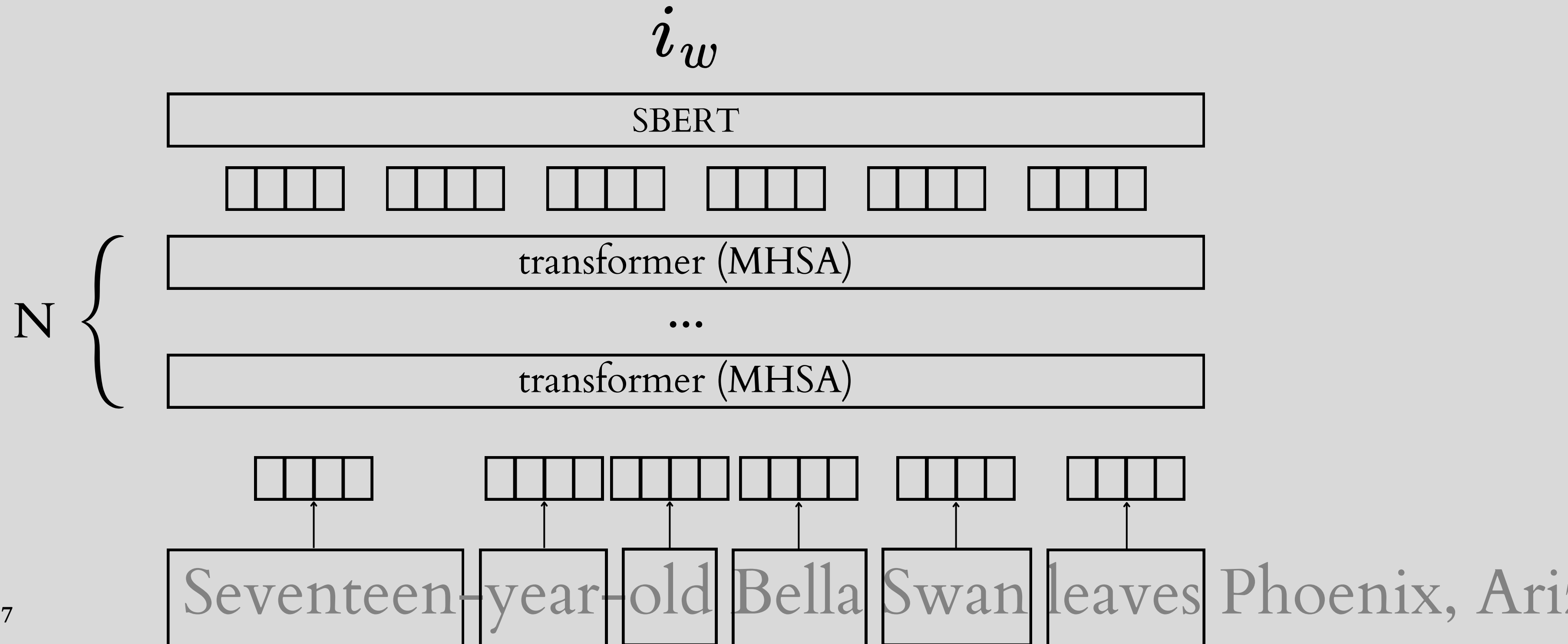
Swan

leaves

Phoenix, Ari

Sentence Encoding

{metodología}



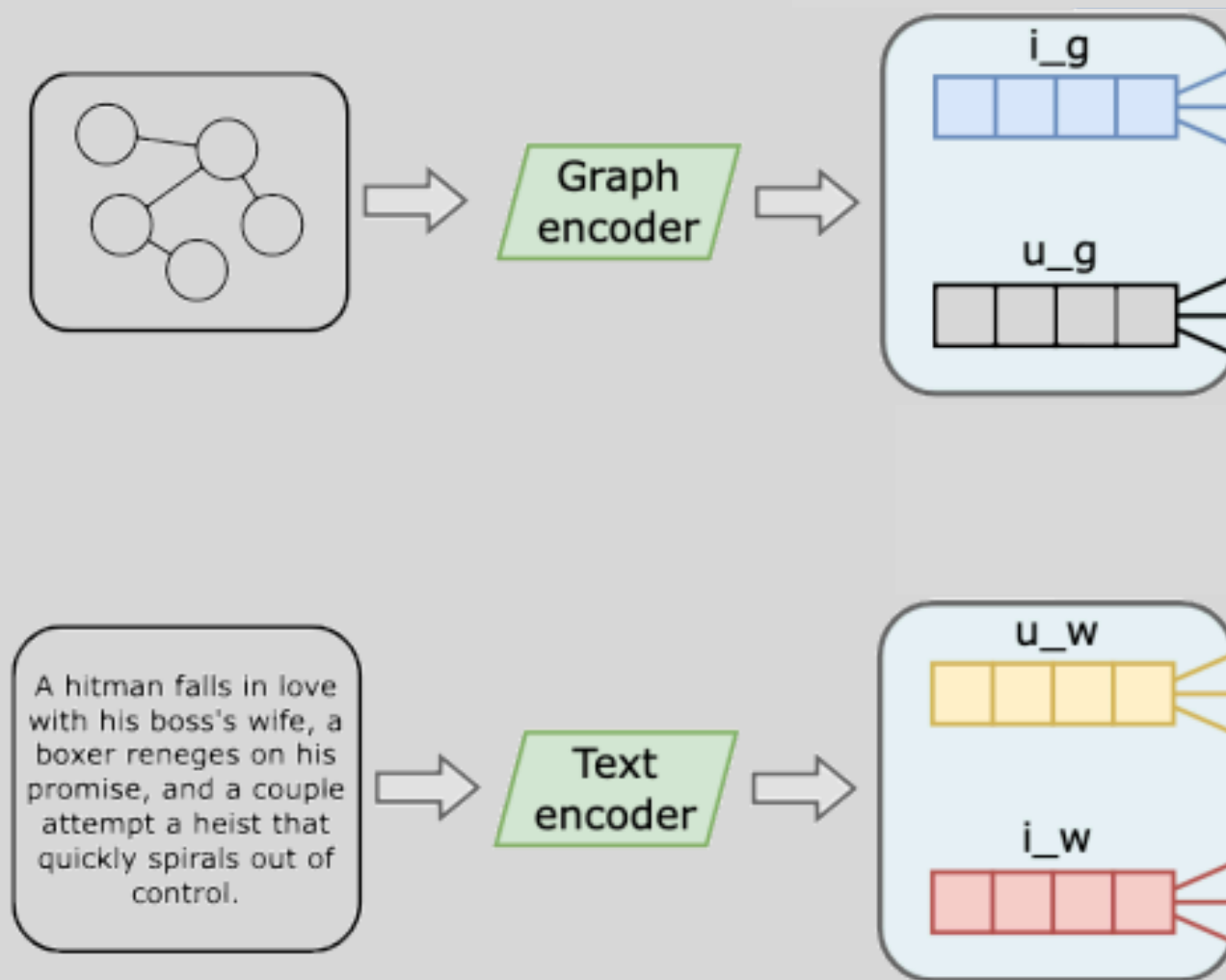
Sentence Encoding

{metodología}

$$\vec{u}_w = \frac{\sum_{i \in L} \vec{i}_w}{|L|}$$

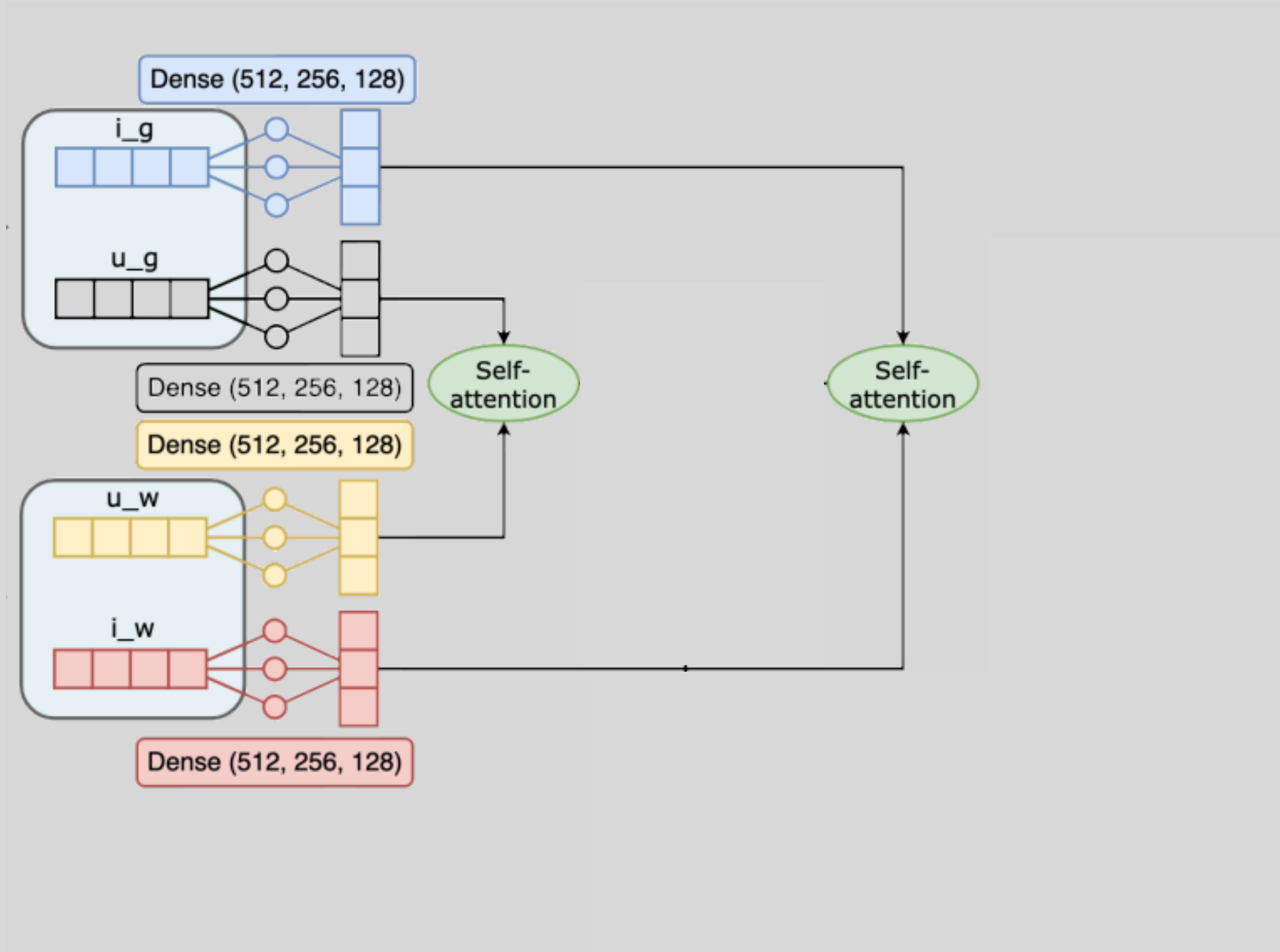
Deep architecture

{metodología}



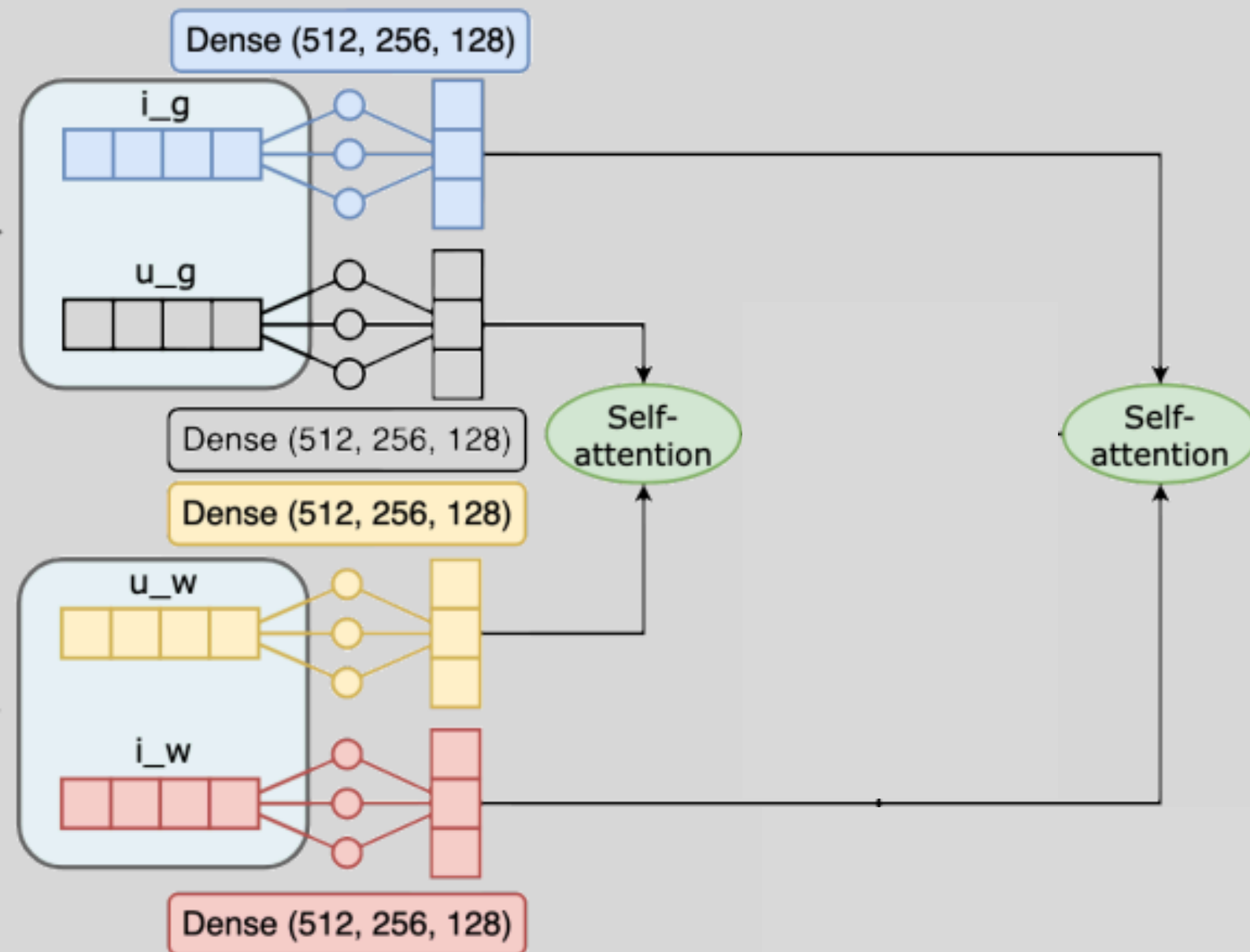
Deep architecture

{metodología}



Deep architecture

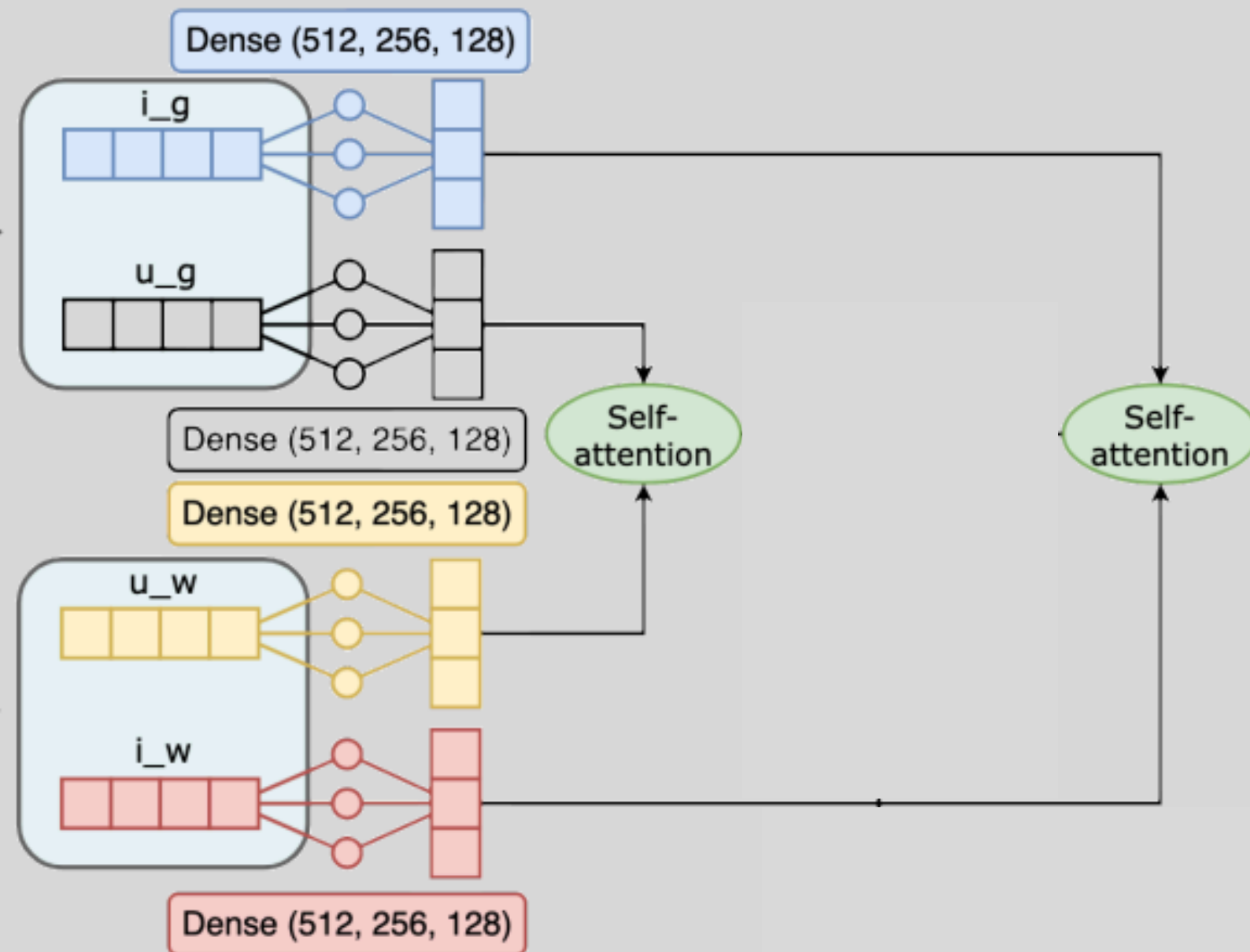
{metodología}



$$\alpha_u = \text{softmax}(W_{att} * (u_g \oplus u_w) + b_{att})$$

Deep architecture

{metodología}

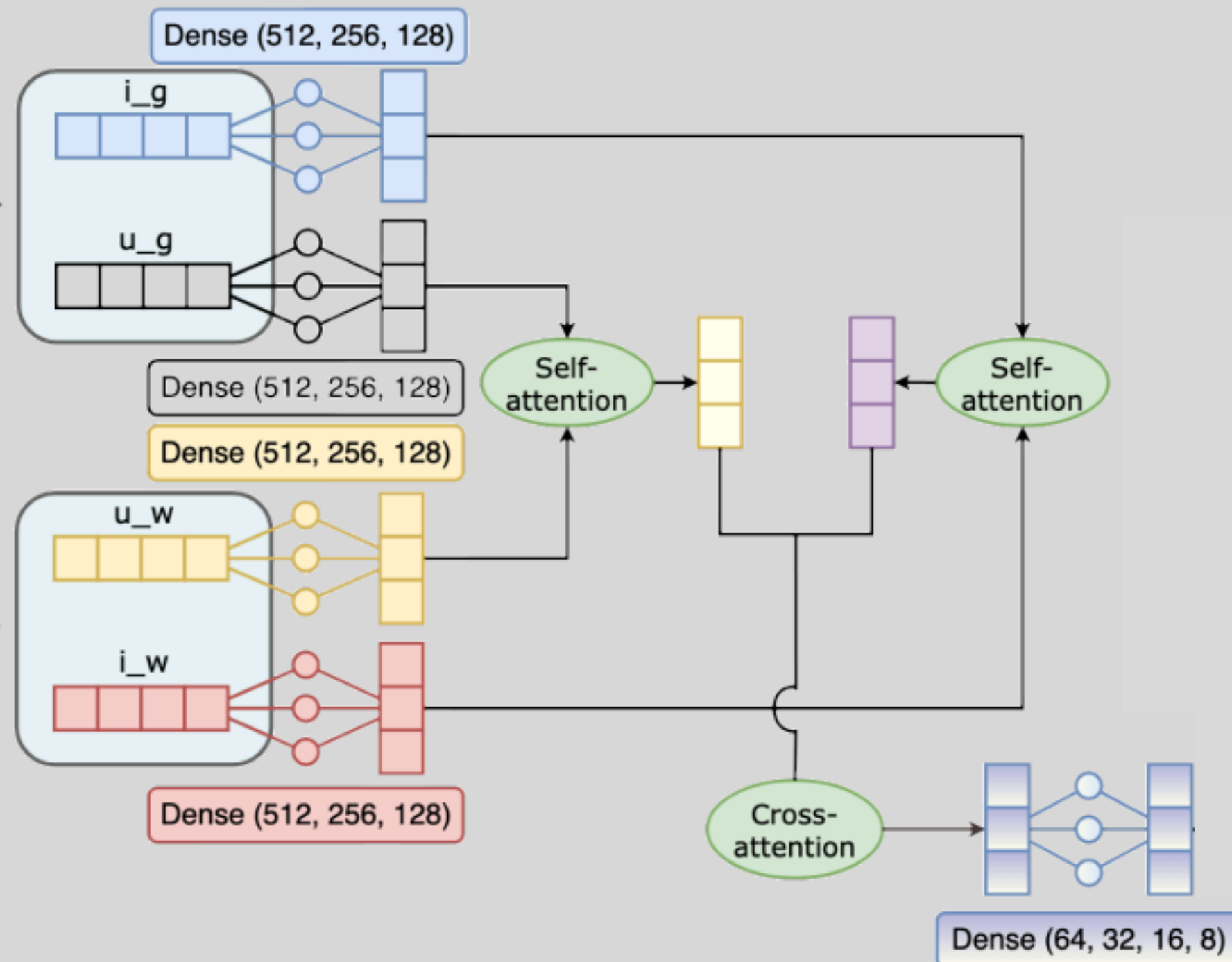


$$\alpha_u = \text{softmax}(W_{att} * (u_g \oplus u_w) + b_{att})$$

$$u = \alpha_u * u_g + (1 - \alpha_u) * u_w$$

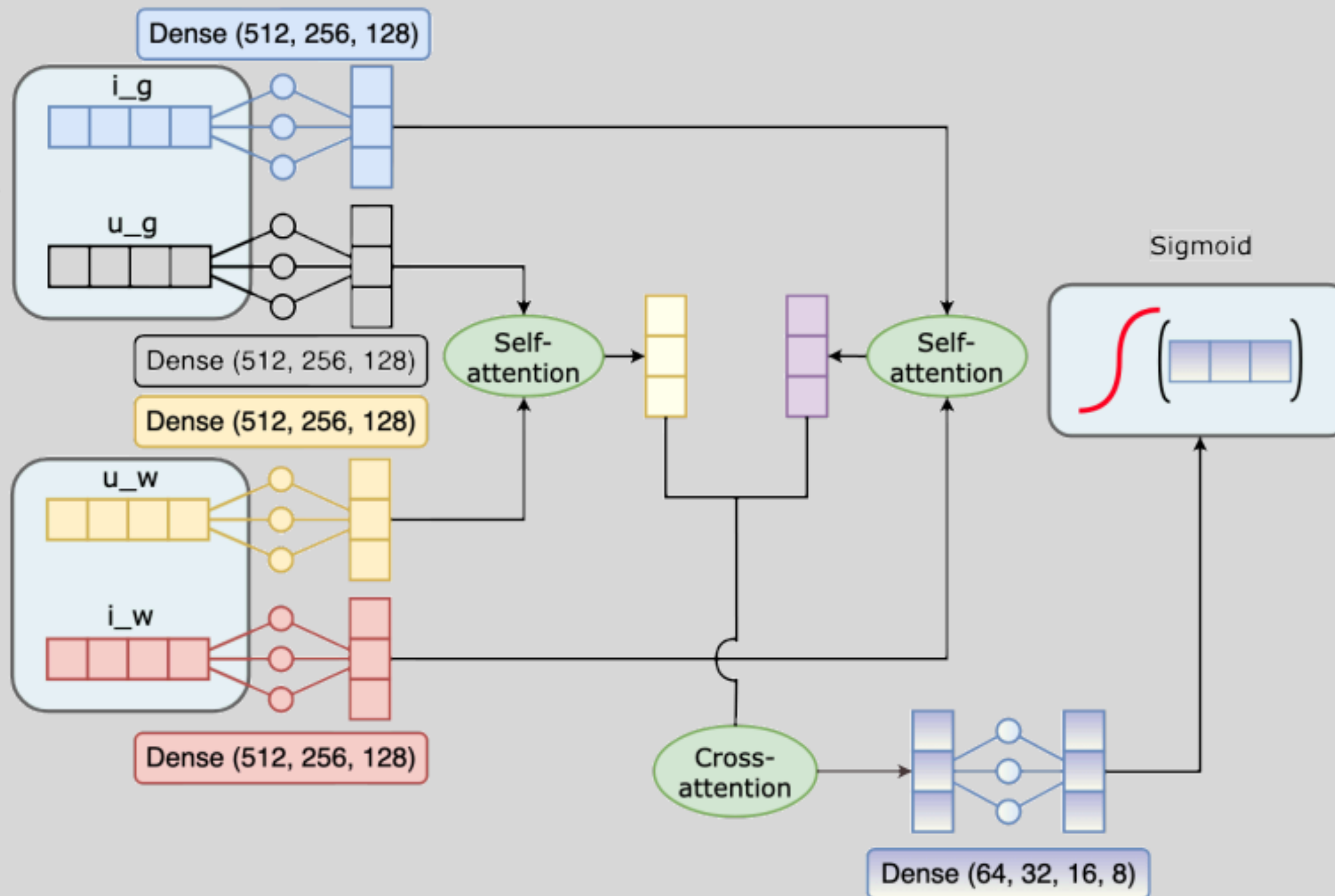
Deep architecture

{metodología}



Deep architecture

{metodología}



{metodología}

$$BCE = -\frac{1}{N} \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

Resultados



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Sistemas Recomendadores
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Resultados

- Contexto

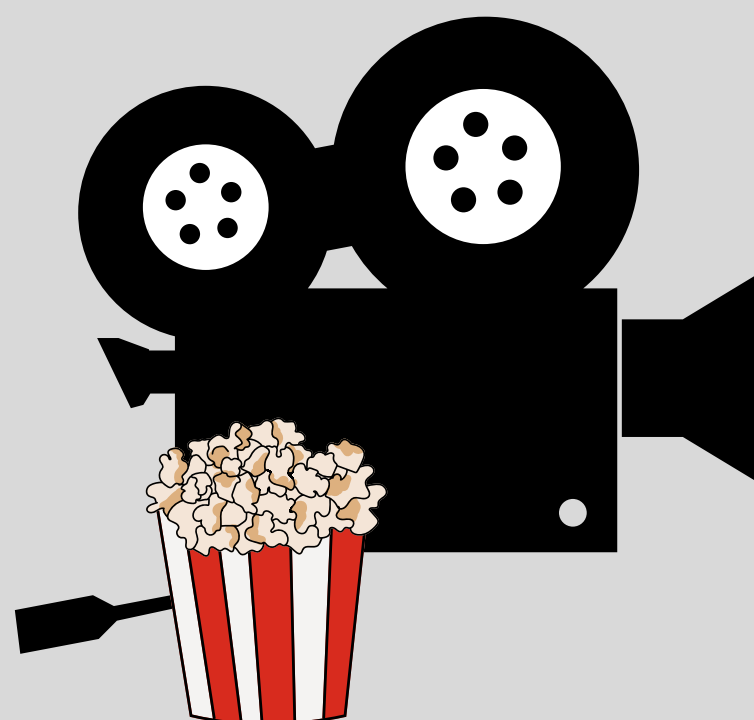


Resultados

- Contexto

	Users	Items	Ratings	%Positive	Sparsity	Avg. ratings/user	Avg. ratings/item	Content feature
ML1M	6,040	3,883	1,000,209	57.51%	96.42%	157.11±182.93	293.98±393.88	plot
DBBOOK	6,181	6,733	72,372	45.85%	99.83%	11.70±5.85	10.74±27.14	abstract

Table 1: Statistics of the datasets.



Resultados

- Contexto

	Users	Items	Ratings	%Positive	Sparsity	Avg. ratings/user	Avg. ratings/item	Content feature
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Table 1: Statistics of the datasets.

	Mapped Items	Entities	Relations	Triples	Avg. Links
ML1M	6,040	26,858	13	828,119	30.83
DBBOOK	6,181	17,505	36	85,549	4.94

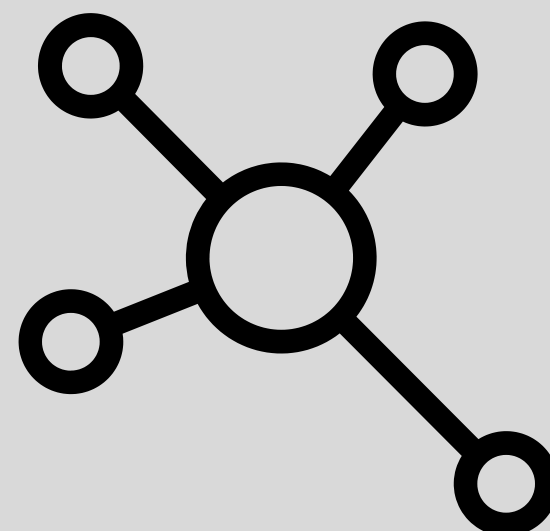
Table 2: Statistics about the Knowledge Graph.

Resultados

- Contexto

	Mapped Items	Entities	Relations	Triples	Avg. Links
ML1M	6,040	26,858	13	828,119	30.83
DBBOOK	6,181	17,505	36	85,549	4.94

Table 2: Statistics about the Knowledge Graph.

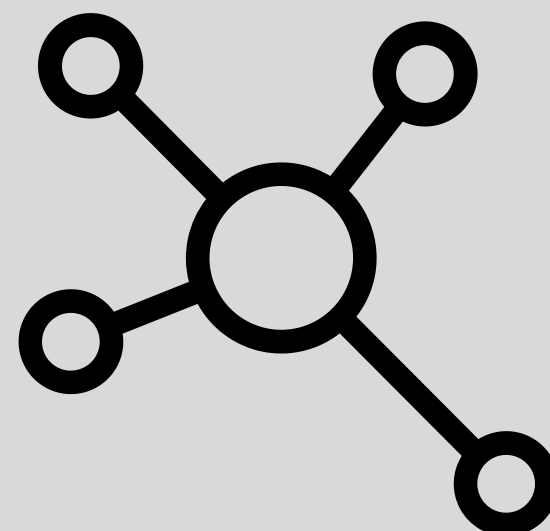


Resultados

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ML1M	6,040	26,858	13	828,119	30.83
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Table 2: Statistics about the Knowledge Graph.



Resultados

Baseline	Accuracy					Diversity	Novelty
	MAP	Precision	Recall	F1	nDCG	Gini	EPC
<i>State-of-the-art Matrix Factorization and Deep Learning Techniques (no KARS)</i>							
BPRMF	0.7486	0.7119	0.4554	0.4636	0.7861	0.0779	0.558
PureSVD	0.7553	0.7121	0.453	0.4615	0.7896	0.113	0.5856
Slim	<i>0.7852</i>	<i>0.7419</i>	0.4612	<i>0.4743</i>	<i>0.8189</i>	0.1897	<i>0.6521</i>
MultiVAE	0.7358	0.6958	0.4483	0.4552	0.7714	0.1445	0.5845
CFGAN	0.6229	0.6127	0.4151	0.414	0.6677	0.2842	0.5396
NGCF	0.5946	0.5929	0.4095	0.4057	0.6442	0.192	0.5229
<i>KARS based on Graph Embeddings and Matrix Factorization Techniques</i>							
AttributeItemKnn	0.7302	0.6911	0.4422	0.449	0.765	0.2257	0.6127
AttributeUserKnn	0.7499	0.714	0.4454	0.4557	0.784	0.1556	0.5957
KaHFM	0.7403	0.7018	0.449	0.4565	0.7763	0.1468	0.5865
CKE	0.7222	0.7057	0.4179	0.4455	0.7535	0.1834	0.5817
CFKG	0.7318	0.7102	0.4193	0.447	0.7604	0.1732	0.5857
KTUP	0.7093	0.6926	0.4117	0.4381	0.7397	0.1743	0.5729
<i>KARS based on Graph Neural Networks and Graph Convolutional Networks</i>							
KGAT	0.7263	0.7064	0.4172	0.445	0.7559	0.1765	0.5812
KGCN	0.7164	0.701	0.4171	0.4439	0.7477	0.1899	0.5814
KGNN-LS	0.7152	0.6998	0.4157	0.4427	0.7462	0.1884	0.5796
LightGCN	0.5946	0.5929	0.4095	0.4057	0.6442	0.192	0.5229
Our best	0.8454**	0.8082**	0.4602	0.5432**	0.8712**	0.1307	0.6657**

Table 3: Comparison to baselines on ML1M data. Best-performing baseline is reported in *italics*. Overall best performing configuration is highlighted in bold. Significant improvements are emphasized with * (for $p < 0.05$) and ** (for $p < 0.01$).

Resultados

Baseline	Accuracy					Diversity	Novelty
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MultiVAE	0.7358	0.6958	0.4483	0.4552	0.7714	0.1445	0.5845
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Resultados

Baseline	Accuracy					Diversity	Novelty
	MAP	Precision	Recall	F1	nDCG	Gini	EPC
<i>State-of-the-art Matrix Factorization and Deep Learning Techniques (no KARS)</i>							
BPRMF	0.6392	0.5721	0.4182	0.4472	0.6372	0.0360	0.6626
PureSVD	0.6365	0.5714	0.4182	0.4468	0.6349	0.0399	0.6622
Slim	0.6240	0.5661	0.4176	0.4437	0.6285	0.0482	0.6556
MultiVAE	0.6222	0.5631	0.4126	0.4400	0.6245	0.0451	0.6519
CFGAN	0.6104	0.5609	0.4124	0.4391	0.6166	0.0507	0.6458
NGCF	0.6074	0.5591	0.4114	0.4377	0.6150	0.0511	0.6448
<i>KARS based on Graph Embeddings and Matrix Factorization Techniques</i>							
AttributeItemKnn	0.6316	0.5674	0.4176	0.4443	0.6333	0.0496	0.6602
AttributeUserKnn	0.6371	0.5700	0.4191	0.4463	0.6363	0.0452	0.6629
KaHFM	0.6384	0.5745	0.4207	0.4493	0.6381	0.0354	0.6633
CKE	0.684	0.6718	0.5203	0.5418	0.7189	0.1309	0.6645
CFKG	0.6844	0.6713	0.5201	0.5421	0.7173	0.1235	0.664
KTUP	<i>0.6845</i>	0.6712	<i>0.5218</i>	0.5424	<i>0.719</i>	0.1041	<i>0.6646</i>
<i>KARS based on Graph Neural Networks and Graph Convolutional Networks</i>							
KGAT	0.682	<i>0.6723</i>	0.5206	<i>0.5427</i>	0.7182	0.1361	0.6661
KGCN	0.6776	0.6664	0.5171	0.5378	0.7131	<i>0.1562</i>	0.6614
KGNN-LS	0.6743	0.668	0.518	0.5389	0.7116	0.1548	0.6604
LightGCN	0.6147	0.5643	0.4156	0.4420	0.6212	0.0462	0.6490
Our best	0.7291**	0.7024**	0.5579**	0.6335**	0.7667**	0.1668*	0.7058

Table 4: Comparison to baselines on DBBOOK data. Best-performing baseline is reported in *italics*. Overall best performing configuration is highlighted in bold. Significant improvements are emphasized with * (for $p < 0.05$) and ** (for $p < 0.01$).

Resultados

Baseline	Accuracy					Diversity	Novelty
	MAP	Precision	Recall	F1	nDCG	Gini	EPC
<i>State-of-the-art Matrix Factorization and Deep Learning Techniques (no KARS)</i>							
BPRMF	0.6392	0.5721	0.4182	0.4472	0.6372	0.0360	0.6626
PureSVD	0.6365	0.5714	0.4182	0.4468	0.6349	0.0399	0.6622
Slim	0.6240	0.5661	0.4176	0.4437	0.6285	0.0482	0.6556
MultiVAE	0.6222	0.5631	0.4126	0.4400	0.6245	0.0451	0.6519
CFGAN	0.6104	0.5609	0.4124	0.4391	0.6166	0.0507	0.6458
NGCF	0.6074	0.5591	0.4114	0.4377	0.6150	0.0511	0.6448
<i>KARS based on Graph Embeddings and Matrix Factorization Techniques</i>							
AttributeItemKnn	0.6316	0.5674	0.4176	0.4443	0.6333	0.0496	0.6602
AttributeUserKnn	0.6371	0.5700	0.4191	0.4463	0.6363	0.0452	0.6629
KaHFM	0.6384	0.5745	0.4207	0.4493	0.6381	0.0354	0.6633
CKE	0.684	0.6718	0.5203	0.5418	0.7189	0.1309	0.6645
CFKG	0.6844	0.6713	0.5201	0.5421	0.7173	0.1235	0.664
KTUP	<i>0.6845</i>	0.6712	<i>0.5218</i>	0.5424	<i>0.719</i>	0.1041	<i>0.6646</i>
<i>KARS based on Graph Neural Networks and Graph Convolutional Networks</i>							
KGAT	0.682	<i>0.6723</i>	0.5206	<i>0.5427</i>	0.7182	0.1361	0.6661
KGCN	0.6776	0.6664	0.5171	0.5378	0.7131	<i>0.1562</i>	0.6614
KGNN-LS	0.6743	0.668	0.518	0.5389	0.7116	0.1548	0.6604
LightGCN	0.6147	0.5643	0.4156	0.4420	0.6212	0.0462	0.6490
Our best	0.7291**	0.7024**	0.5579**	0.6335**	0.7667**	0.1668*	0.7058

Table 4: Comparison to baselines on DBBOOK data. Best-performing baseline is reported in *italics*. Overall best performing configuration is highlighted in bold. Significant improvements are emphasized with * (for $p < 0.05$) and ** (for $p < 0.01$).

Resultados

Configuration	Accuracy					Diversity	Novelty
	MAP	Precision	Recall	F1	nDCG	Gini	EPC
Our best	0.8454	0.8082	0.4602	0.5432	0.8712	0.1307	0.6657
<i>Architecture</i>							
w/o graph	0.8397 (*)	0.8031 (**)	0.4587	0.541	0.865 (*)	0.1437	0.6671 (**)
w/o text	0.8415	0.8058	0.4587	0.5415	0.8675	0.1474	0.6675
w/o attention	0.8334 (*)	0.7996 (**)	0.4576	0.5395	0.8605 (*)	0.1333	0.6595
w/o dropout	0.8394 (*)	0.805	0.4582	0.5408	0.8661	0.1447	0.6687
<i>Features</i>							
w/o properties	0.8429	0.8048 (*)	0.4587	0.5413	0.8682	0.1265	0.6632
w/o prop & text	0.8368	0.8043 (*)	0.4583	0.541	0.8639	0.1600 (*)	0.6715 (*)

Table 5: Results of the ablation test on ML1M data. Best-performing configuration is reported in bold. Each line indicates the module which is removed from the architecture. Significant improvements are emphasized with (*) (for $p < 0.05$) and (**) (for $p < 0.01$).

Configuration	Accuracy					Diversity	Novelty
	MAP	Precision	Recall	F1	nDCG	Gini	EPC
Our best	0.7291	0.7024	0.5579	0.6335	0.7667	0.1668	0.7058
<i>Architecture</i>							
w/o graph	0.6968 (**)	0.6842 (**)	0.5416 (**)	0.6151 (**)	0.7373 (**)	0.2481	0.6822 (**)
w/o text	0.7022 (**)	0.6883 (**)	0.5463 (**)	0.6204 (**)	0.7418 (**)	0.2261	0.6867 (**)
w/o attention	0.7052 (**)	0.6909 (**)	0.5479 (**)	0.6239 (**)	0.7459 (**)	0.2293	0.6900 (**)
w/o dropout	0.7042 (**)	0.6916 (**)	0.5502 (*)	0.6242 (*)	0.7464 (**)	0.2306	0.6900 (**)
<i>Features</i>							
w/o properties	0.7231 (**)	0.6975 (**)	0.5565 (*)	0.6314 (*)	0.7619	0.1935	0.7019 (**)
w/o prop & text	0.6875 (**)	0.6814 (**)	0.5409 (**)	0.6145 (**)	0.7325 (**)	0.2410	0.6767 (**)

Table 6: Results of the ablation test on DBBOOK. Best-performing configuration is reported in bold. Each line indicates the module which is removed from the architecture. Significant improvements are emphasized with (*) (for $p < 0.05$) and (**) (for $p < 0.01$).

Resultados

Configuration	Accuracy					Diversity	Novelty
	MAP	Precision	Recall	F1	nDCG	Gini	EPC
Our best	0.8454	0.8082	0.4602	0.5432	0.8712	0.1307	0.6657
<i>Architecture</i>							
w/o graph	0.8397 (*)	0.8031 (**)	0.4587	0.541	0.865 (*)	0.1437	0.6671 (**)
w/o text	0.8415	0.8058	0.4587	0.5415	0.8675	0.1474	0.6675
w/o attention	0.8334 (*)	0.7996 (**)	0.4576	0.5395	0.8605 (*)	0.1333	0.6595
w/o dropout	0.8394 (*)	0.805	0.4582	0.5408	0.8661	0.1447	0.6687
<i>Features</i>							
w/o properties	0.8429	0.8048 (*)	0.4587	0.5413	0.8682	0.1265	0.6632
w/o prop & text	0.8368	0.8043 (*)	0.4583	0.541	0.8639	0.1600 (*)	0.6715 (*)

Table 5: Results of the ablation test on ML1M data. Best-performing configuration is reported in bold. Each line indicates the module which is removed from the architecture. Significant improvements are emphasized with (*) (for $p < 0.05$) and (**) (for $p < 0.01$).

Configuration	Accuracy					Diversity	Novelty
	MAP	Precision	Recall	F1	nDCG	Gini	EPC
Our best	0.7291	0.7024	0.5579	0.6335	0.7667	0.1668	0.7058
<i>Architecture</i>							
w/o graph	0.6968 (**)	0.6842 (**)	0.5416 (**)	0.6151 (**)	0.7373 (**)	0.2481	0.6822 (**)
w/o text	0.7022 (**)	0.6883 (**)	0.5463 (**)	0.6204 (**)	0.7418 (**)	0.2261	0.6867 (**)
w/o attention	0.7052 (**)	0.6909 (**)	0.5479 (**)	0.6239 (**)	0.7459 (**)	0.2293	0.6900 (**)
w/o dropout	0.7042 (**)	0.6916 (**)	0.5502 (*)	0.6242 (*)	0.7464 (**)	0.2306	0.6900 (**)
<i>Features</i>							
w/o properties	0.7231 (**)	0.6975 (**)	0.5565 (*)	0.6314 (*)	0.7619	0.1935	0.7019 (**)
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Table 6: Results of the ablation test on DBBOOK. Best-performing configuration is reported in bold. Each line indicates the module which is removed from the architecture. Significant improvements are emphasized with (*) (for $p < 0.05$) and (**) (for $p < 0.01$).

Referencias Principales

- Spillo, G., Musto, C., Polignano, M., Lops, P., de Gemmis, M., & Semeraro, G. (2023). Combining graph neural networks and sentence encoders for knowledge-aware recommendations. UMAP '23: Proceedings of the 31st ACM Conference on User Modeling, Adaptation and Personalization.

Fin

MUCHAS GRACIAS POR SU ATENCIÓN